Statics and equilibrium

Introduction

Mechanics could be defined as the science of predicting the conditions of rest or motion of bodies under the action of **forces.**

Statics is that branch of **mechanics** that deals with systems maintaining a rigid body in a state of **equilibrium**. **Statics** analysis studies the **forces** acting on or within bodies at rest or in a steady motion.

Force is a **vecto**r physical quantity, which tends to change the state of equilibrium of a rigid body. The force can be applied directly to a body or indirectly as in the case of the earth's gravity.

Type of forces:

Surface force

Concentrated force

Distributed load

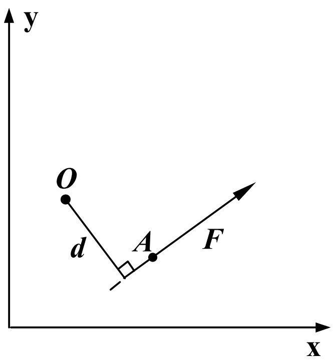
Body force

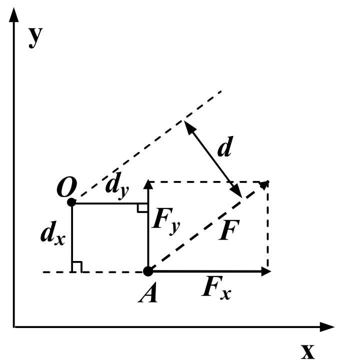
Support reactions

Moment is the tendency of an applied force to produce a rotation about any axis that does not intersect its line of action. A moment is also a vector quantity.

The product of the **force F** and the **perpendicular distance to any point, d,** give the magnitude of the moment from a point to the force line of action:

$$\boxed{M = F \times d}$$



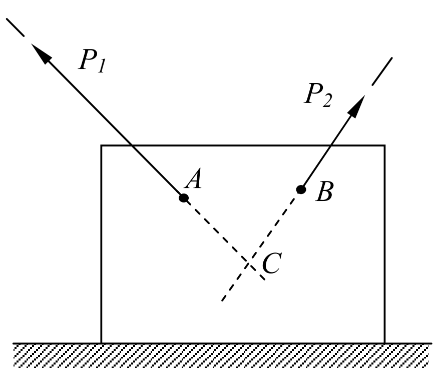


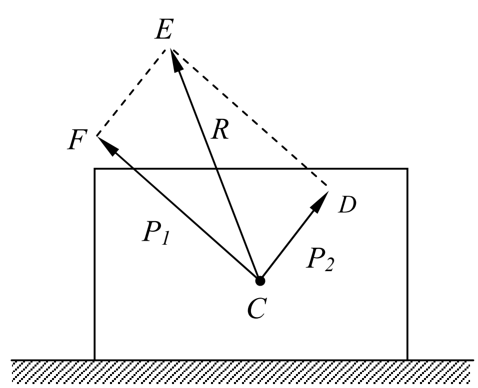
Force Units
[N] (Newton) – "A force of one Newton must be applied to a mass of one kilogram in order to accelerate it with an acceleration of one meter per second squared"

Moment has units of: Newton metre = Nm

Resultant of forces:

When the line of action of two forces P_1 and P_2 intersect, their combined resultant could be represented by the diagonal of a ***parallelogram*** whose sides has the magnitude and direction of the two forces:



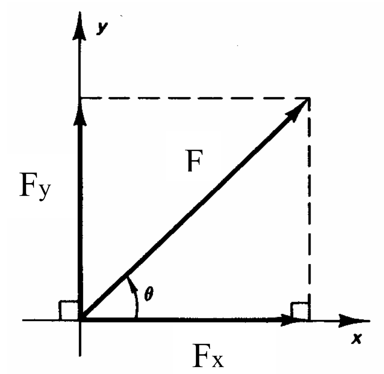


Force components:

It is also possible to divide an external force into two forces - called the force components - using the same principles of the parallelogram above. The components F_x and F_y that act along the x and y-axes have the direction and magnitude given by:

$$F_y = F \sin\theta \qquad F = \sqrt{F_x^2 + F_y^2}$$

$$F_x = F \cos\theta \qquad \theta = \tan^{-1}\frac{F_y}{F_x}$$



Equilibrium conditions

A rigid body is said to be in ***equilibrium*** if the resultant of all forces and moments acting on it is zero. A body is in ***static equilibrium*** if it shows no tendency to move in any direction or to rotate about any axes.

These conditions can be expressed mathematically by the two vector equations:

$$\Sigma F = 0$$

$$\Sigma M = 0$$

Where ΣF represents the sum of all the forces acting on the body and ΣM is the sum of the moments of all the forces about any point on or off the body.

The force and moments can be resolved into components along the coordinate axes and the above equations can be written as six scalar equations:

$$\Sigma F_x = 0 \qquad \Sigma F_y = 0 \qquad \Sigma F_z = 0$$

$$\Sigma M_x = 0 \qquad \Sigma M_y = 0 \qquad \Sigma M_z = 0$$

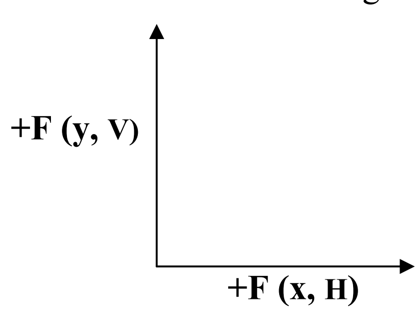
Often in engineering practice, a plan system is considered and the equilibrium conditions can be represented using three equations:

$$\Sigma F_x = 0 \qquad \Sigma F_y = 0 \qquad \Sigma M = 0$$

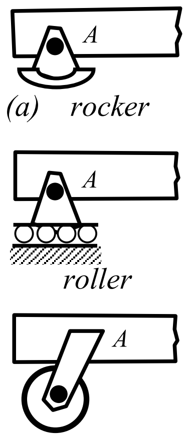
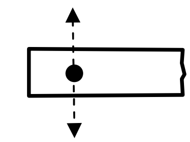
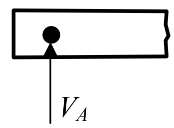
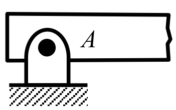
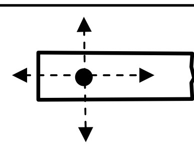
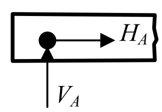
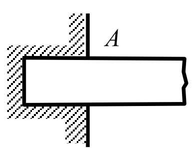
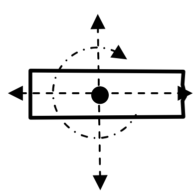
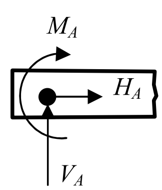
In this case the moment can be taken about any point on the x-y plane.

The free body diagram and support reactions

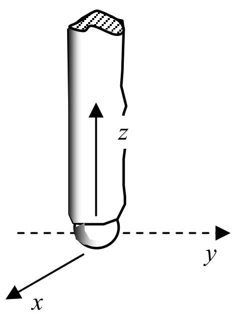
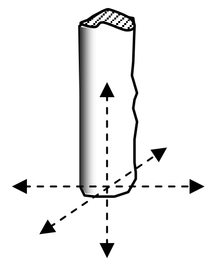
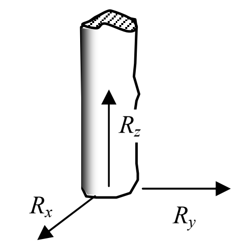
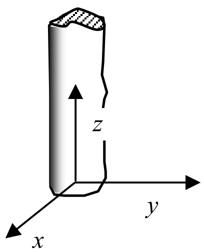
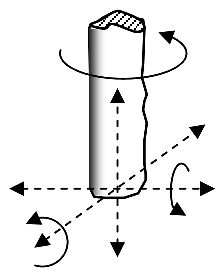
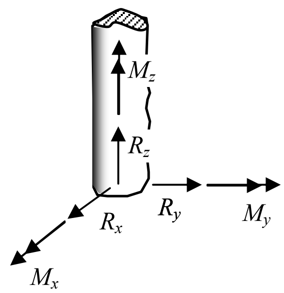
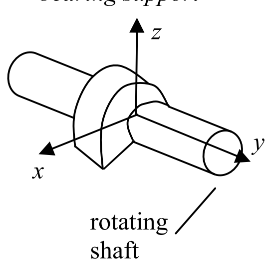
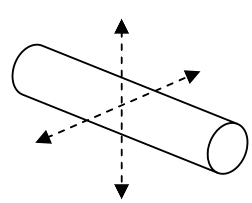
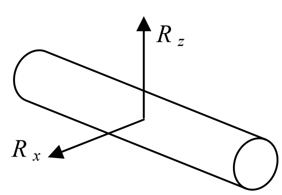
A free body diagram (FBD) is a diagram of an object, or part of an object, on which all the external forces are indicated. Drawing a correct FBD is an important step in the static analysis. To draw the FBD of the structure or part, the support conditions need to be replaced by external forces. These forces are usually unknown and the type of the support indicates the type of external forces and line of action to consider. We consider upwards vertical forces and from the left to right horizontal forces to have a positive (+) sign:



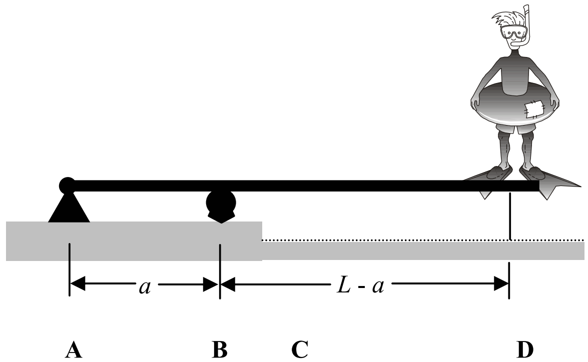
Type of supports for a planar system:

Type of support	Restraint Offered	Equivalent Forces
 *(a) rocker*, *roller*, *frictionless wheel*		 V_A $H_A = 0$ $M_A = 0$
 *(b) frictionless pin (i.e., hinge)*		 H_A, V_A $M_A = 0$
 *(c) fixed end (built in support)*		 M_A, H_A, V_A

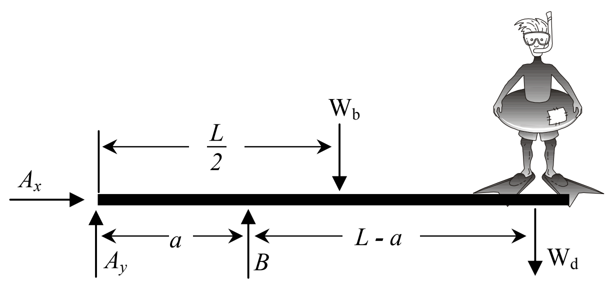
Type of supports for space structure:

Type of support	Restraint Offered	Equivalent Forces
(a) ball – and - socket 	*prevents translation but allows rotation* 	 $M_x = M_y = M_z = 0$
(b) fixed end (built-in support) 	*prevents translation and rotation* 	
(c) frictionless collar bearing support 		 $M_y = 0, \quad R_y = 0$

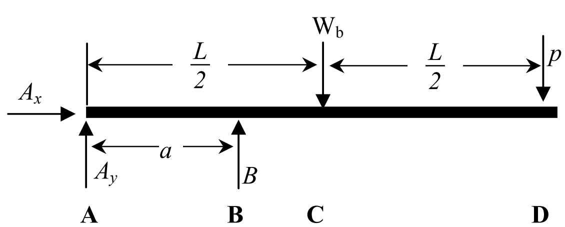
Example 1



a. Diver on a diving board



b. FBD of diver and board

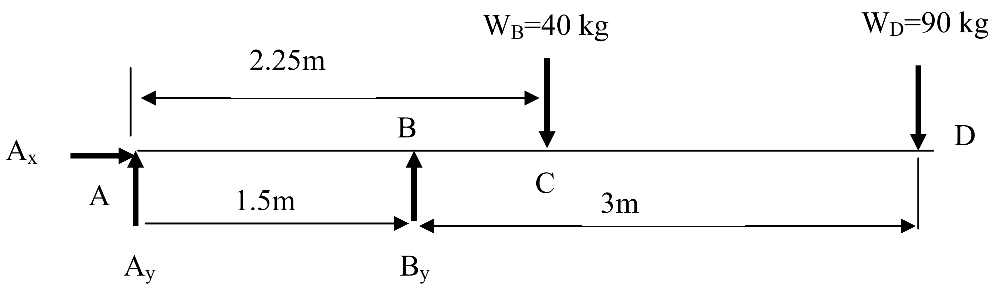


c. FBD of board

If a = 1.5m, L = 4.5m and the respective weights of the diver and the board are 90kg and 40kg, find the reactions at the pin A and the roller B.

Solution:

Free Body Diagram:



$[\Sigma F_x = 0]$ $\quad A_x = 0$

$[\Sigma M_A = 0]$ $\quad 1.5 \text{ x } B_y – (2.25 \text{ x } 40 \text{ x } 9.81) – (4.5 \text{ x } 90 \text{ x } 9.81) = 0$

$1.5 \text{ x } B_y – 4854 = 0$

ACW $\quad B_y = 3.236 \text{ kN}$

$[\Sigma F_y = 0]$ $\quad A_y + B_y – W_B - W_D = 0$

$Ay = 40 \text{ x } 9.81 + 90 \text{ x } 9.81 –3236 = - 1.96 \text{ kN}$

Check using moment at C:

$[\Sigma M_C = 0]$

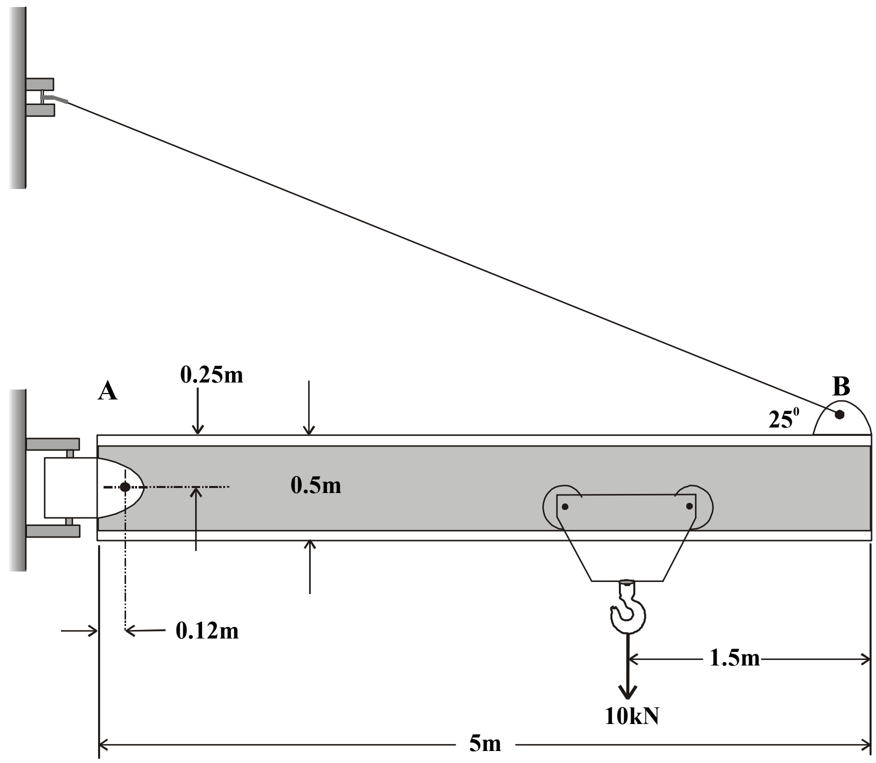
$-1.96 \text{ x } 2.25 + 3.24 \text{ x } 0.75 + 90 \text{ x } 9.81 \text{ x } 2.25 \text{ x } 10^{-3} = >$

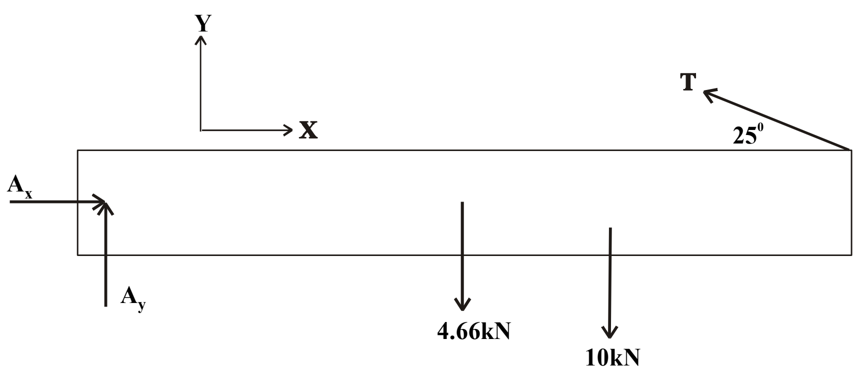
$= – 4.41 + 2.427 + 1.985 = 0$

CW

Example 2

Determine the tension force in the supporting cable and the reaction force on the pin A for the jib crane shown. The beam is a standard 0.5m I beam with a mass of 95 kg/m.





Free body diagram of the jib

Solution:

The weight of the beam is: 95[kg] x 9.81[ms^{-2}] x 5[m] = 4.66kN. We assume the beam weight acts in the middle of the beam.

There are 3 unknown forces: T, A_x and A_y, which could be found by using the three equations of equilibrium for plane structure.

Applying the moment equation about point A eliminates two of the problem unknowns. It is also simpler to consider the moment using the x and y components of T than it is to compute the perpendicular distance from T to A. The moment taken is positive in the counterclockwise direction:

$[\Sigma M_a = 0]$ (+)

$$(T \cos 25^0)0.25 + (T \sin 25^0)(5-0.12) - 10(5 - 1.5 - 0.12) - 4.66(2.5 - 0.12) = 0$$

$$T = 19.61\text{kN}$$

$[\Sigma F_x = 0]$

$$A_x - 19.61 \cos 25^0 = 0$$

$$A_x = 17.77\text{kN}$$

$[\Sigma F_y = 0]$

$$A_y + 19.61 \sin 25^0 - 4.66 - 10 = 0$$

$$A_y = 6.37\text{kN}$$

And the magnitude and direction of the support at A is:

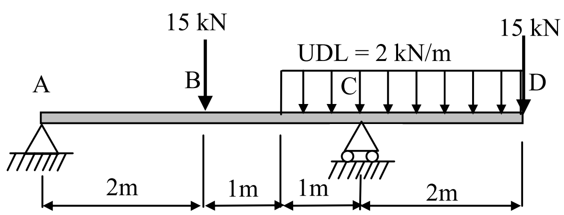
$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{17.77^2 + 6.37^2}$$
$$A = 18.88\text{kN}$$

$$\theta = \tan^{-1}\frac{A_y}{A_x} = \tan^{-1}\frac{6.37}{17.77}$$
$$\theta = 19.7^{\circ}$$

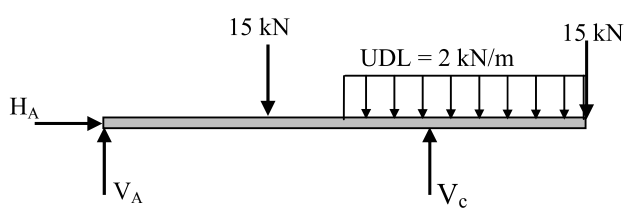
Example 3:

A beam is pin supported at A and roller supported at C and carries two concentrated loads of 15 kN and a uniformly distributed load (UDL) of 2 kN/m as shown. Determine the reaction forces.

Loading Diagram



Free Body Diagram



Solution

$\Sigma H=0$ $\qquad$ $\mathbf{H_A=0}$

To determine V_C take moment about A:

$\Sigma M_A=0$

$+(15x2) - (V_C x4) + (2x3x4.5) + (15x6) = 0$ $\qquad$ $\mathbf{V_C = 36.75\ kN}$

To determine V_A:

$\Sigma V=0$

$V_A - 15 + V_C - (2x3) - 15 = 0$ $\qquad$ $\mathbf{V_A = -0.75\ kN}$

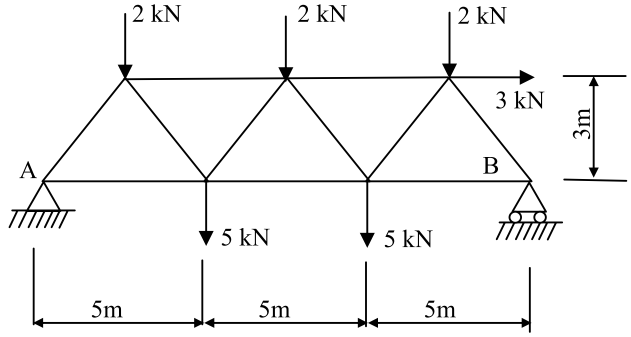
Checking by taking moments about C:

$\Sigma M_C = V_A X4 - 15x2 + 2x3x0.5 + 15x2 = (-0.75x4)-30+3+30 = 0$

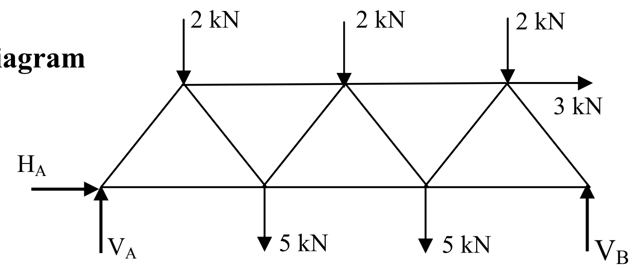
Example 4

Determine the reaction forces at the supports for the truss below when the loads act as shown.

Loading Diagram



Free Body Diagram



Solution

$\Sigma H=0 \qquad H_A + 3 = 0 \qquad H_A = -3$ kN

$\Sigma M_A=0$

$(2x2.5) + (5x5) + (2x7.5) + (5x10) + (2x12.5) + (3x3) – (V_Bx15) = 0$

$\mathbf{V_B = 8.6}$ **kN**

$\Sigma V=0$

$V_A – 2 - 5 –2 – 5 – 2 + V_B = 0 \qquad$ $\mathbf{V_A = 7.4}$ **kN**

Checking by taking moments about B:

$\Sigma M_B = V_AX15 – 2x12.5 - 5x10 - 2x7.5 – 5x5 – 2x2.5 + 3x3 = 0$

Review of the main steps for drawing a FBD:

Step1: Decide which free body is desired and isolate it from the rest of the structure.

Step 2: Show all externally applied forces and moments.

Step 3: Include the weight of the body if applicable.

Step 4: Replace the supports with the appropriate reaction forces.

Summary:

This section covered some basic principles in equilibrium and static mechanics. Different types of forces and moments and how to determine the resultant of several given forces were described. The importance of using a free body diagram in static analysis was demonstrated and the different types of support reactions were shown.

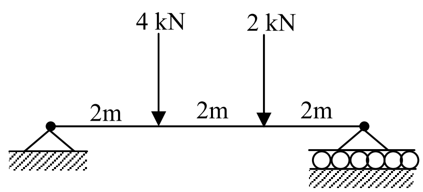
Topics for further reading:

Space structures
Vectors analysis
Determinate and indeterminate conditions
Friction

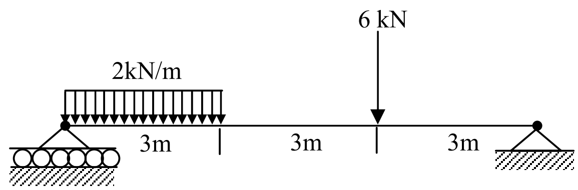
Tutorials

For the following structures calculate the reaction forces at the supports. Assume equilibrium conditions and negligible structure weight.

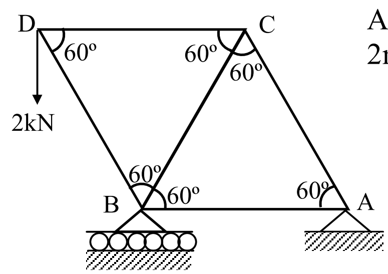
1.



2.

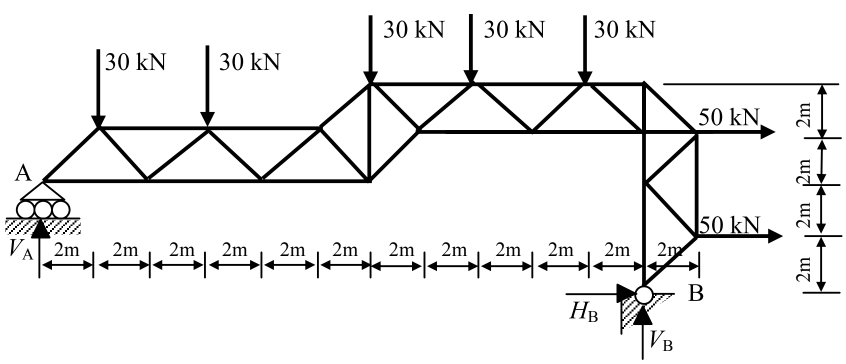


3.

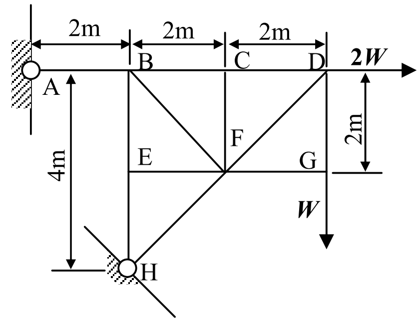


All members are 2m long

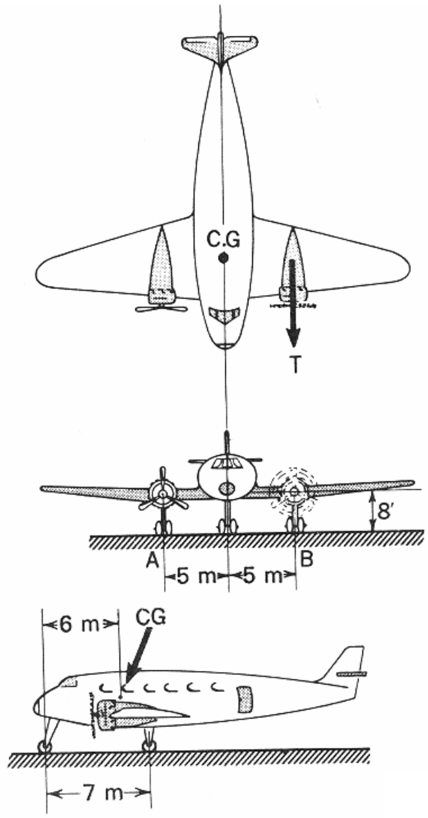
4.



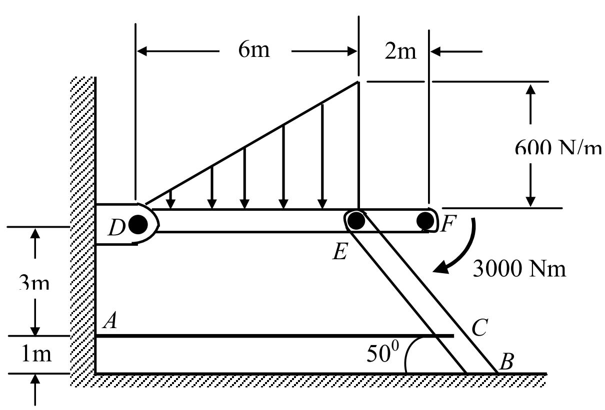
5.



6. A transport plane is being engine tested on the ground prior to take off as shown below. Wheels A and B are locked by the braking system while one engine is developing a thrust T of 15kN. If the total weight of the plane is 30000 kg what are the reaction forces on the wheels?



7. A frame carries a distributed load, which varies linearly from 0 to 600 N/m over a 6m length of the member DEF. A clockwise moment of 3000 Nm is applied to the free end of DEF at F. Determine the tension in the wire AC and the reaction forces at B and D.



Framework Structures

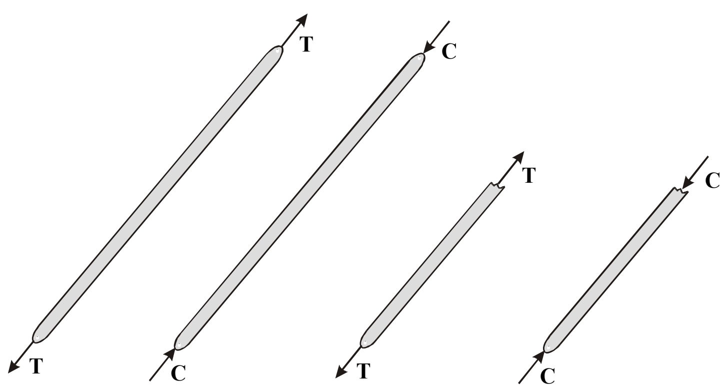
Introduction

In our first lecture we focused on the analysis of a single rigid body using the equilibrium equations to determine all external forces and moments that act on a body. In this section we concentrate on the determination of the internal forces in structures. These structures are typically trusses and frames and often use a pin type of joint to connect the structural members.

We shall limit our discussion to statically determinate structures that do not have more supporting constraints than are necessary to maintain equilibrium. In principle, the same equilibrium equations developed in the previous lecture will be applied to the framework structures analysis to obtain the internal forces.

Definitions and assumptions

A frame or truss is an idealised structure, which consists of straight bars (members), each of which is pinned to the rest of the structure and/or to the ground at both end points. The member is assumed to carry a load only in its longitudinal direction. To satisfy the equilibrium, each member is therefore under two equal, opposite and collinear forces. The member can be in tension or compression as shown in the figure below.



A frame that is constructed in a plane is called a ***plane structure*** while a ***space structure*** is three dimensional. Usually the member's weight is considered negligible in the analysis.

Two methods for analysing the internal forces in simple frames are described and demonstrated using a simple framework. The two methods are often referred to as the ***joint method*** and the ***section method***. In practice, one can 'mix and match' the two methods to work out the internal forces. However, in most cases, in the first stage, a FBD of the whole structure is required to find the external forces by using the equilibrium equations before analysis of the internal forces begins.

Statically indeterminate truss and redundancies

A framework or truss can be ***externally indeterminate*** or ***internally indeterminate*** or both. If, in a frame, the number of external supports is more than necessary to ensure a stable equilibrium state, the extra supports are ***external redundancies*** and the frame as a whole is statically indeterminate. If the number of internal members in a framework is larger than required to prevent collapse, the frame has ***internal redundancies.***

For a statically determinate truss there are definite relations between the number of the members and the number of joints required for the internal stability. For a plane structure each joint has two force equations and therefore there are **2j** equations for the whole simple truss with j joints. If a frame is ***externally determinate*** with three unknown supports and composed of m internal members the total numbers of unknowns in the structure are **m+3**. Thus, for statically determinate structures composed of *triangular* elements, the relation between the number of joints and the number of unknowns is:

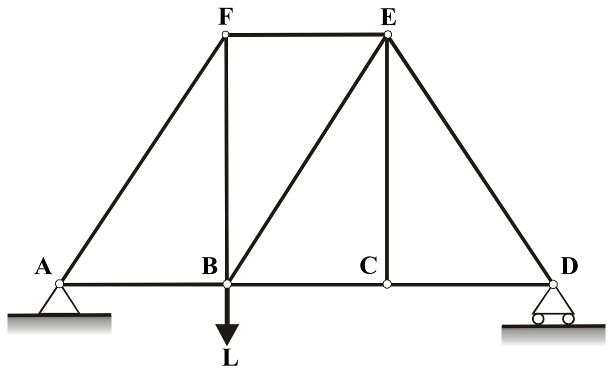
$$m + 3 = 2j$$

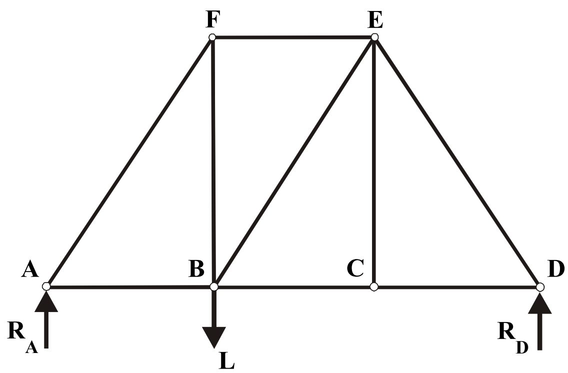
If $m + 3 > 2j$ there are more members than equations and the frame is **statically indeterminate** with internal redundant members. If $m + 3 < 2j$ there are less members than independent equations and the truss is unstable and would collapse under load. This discussion may be extended to space structures where an internal statically determinate frame will satisfy the relation:

$$m + 6 = 3j$$

Method of joints

Consider the simple frame below. A FBD is constructed and the external forces, R_A and R_D in the supports are calculated using the equilibrium equations before the internal force analysis begins.

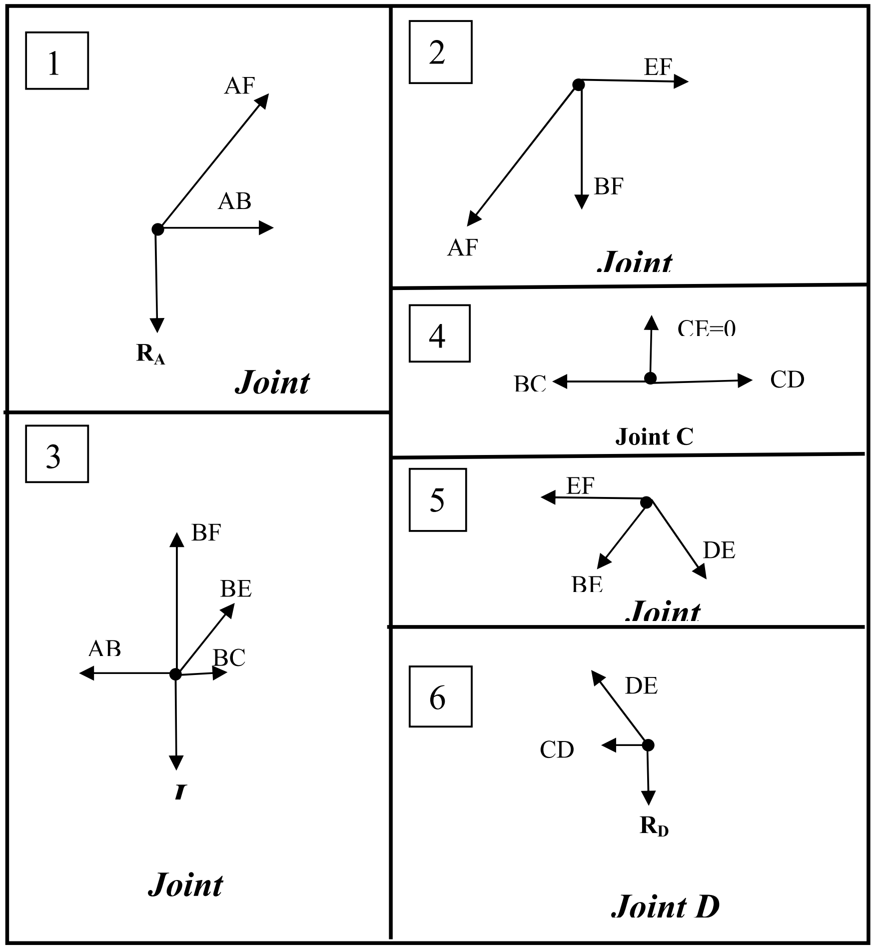




The forces acting on the connecting pin of each joint must be in equilibrium. The analysis is applied for each joint separately where, for plane truss two equilibrium equations, the sum of the horizontal forces, $\mathbf{\Sigma F_x = 0}$, and the sum of the vertical forces, $\mathbf{\Sigma F_y = 0}$, must be satisfied.

Therefore, the analysis may only be applied to joints ***with two unknown forces***. The solution for each pin or joint starts with a free body diagram of the joint indicating all the external and internal forces acting on the joint inspected. The internal member forces are always ***in the direction of the member*** (why?). If a resultant member force acts away from the joint the member is in tension while if the force acts into the joint the member is in compression (principle of action-reaction mechanism).

Analysis of a simple frame internal joint forces:

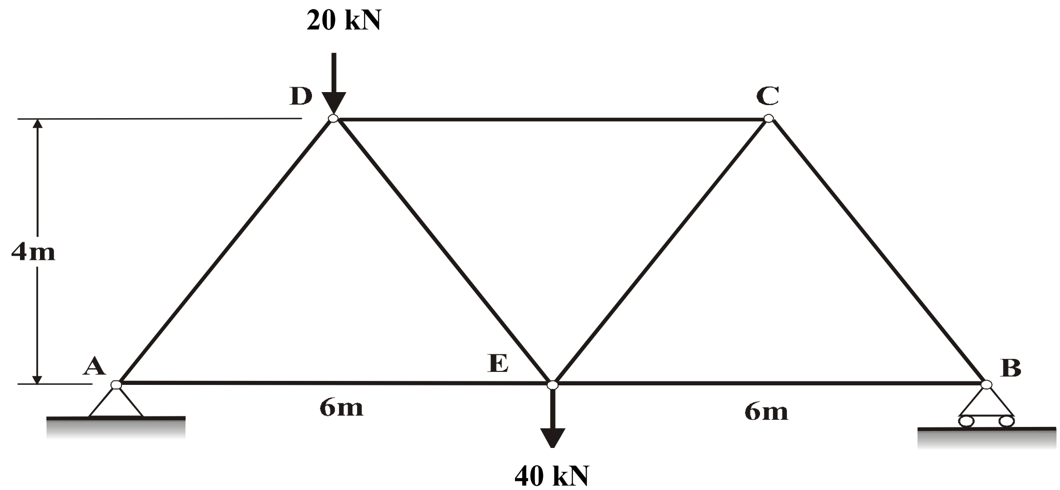


An example of sequential joints analysis used to obtain the internal force members is shown in the figure above. Note that in the joint FBD of each step we have two unknown forces while the other forces were known from the previous steps.

- Some members are in compression and some in tension (which?).
- The force magnitude and direction is the same at any point on a particular member.
- The force CE = 0 (why?).

Example 1 (method of joints)

The framework shown below carries two forces of 20kN at D and 40kN at E. The frame is roller supported at B and pin supported at A. Determine all the internal forces and directions (tension or compression) in the frame.



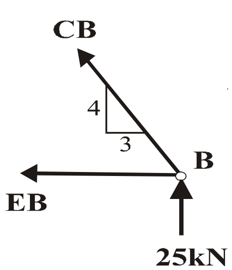
Solution:

We begin by finding the reactions at A and B using a FBD of the all structure. We obtain:

$$R_A = 35\text{kN} \text{ and } R_B = 25\text{kN} \text{ (check this yourself).}$$

Now we can apply the method of joints. We start at joint B where only two forces are unknown and work our way to joints C and D. Can you find a different joint to start? Note that we assume each member is in TENSION (forces acting away from the joint). A member is in compression if the numerical force result is negative.

Joint B

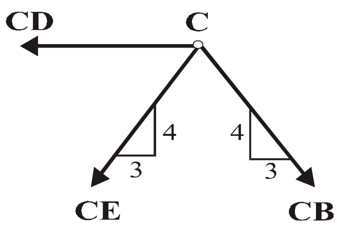


$$\Sigma F_y = 0 \qquad \frac{4}{5}CB + 25 = 0$$

$$CB = -31.25\text{kN} \quad \text{(compression)}$$

$$\Sigma F_x = 0 \qquad EB + \frac{3}{5}CB = 0$$
$$EB = 18.75\text{kN} \quad \text{(tension)}$$

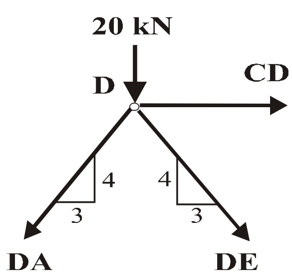
Joint C



$$\Sigma F_y = 0 \qquad \frac{4}{5}CE + \frac{4}{5}CB = 0$$
$$CE = 31.25\text{kN} \quad \text{(tension)}$$

$$\Sigma F_x = 0 \qquad \frac{3}{5}CB - \frac{3}{5}CE - CD = 0$$
$$CD = -37.5\text{kN} \quad \text{(compression)}$$

Joint D



$$\Sigma F_y = 0 \qquad 20 + \frac{4}{5}DE + \frac{4}{5}DA = 0$$

$$\Sigma F_x = 0 \qquad CD + \frac{3}{5}DE - \frac{3}{5}DA = 0$$

We have two equations and two unknowns. Elimination procedure:

$$\left.\begin{aligned} DA + DE &= -25 \\ DE - DA &= 62.5 \end{aligned}\right\}$$

$$DE = 18.75\text{kN} \quad \text{(tension)}$$
$$DA = -43.75\text{kN} \quad \text{(compression)}$$

Check:

$$20 + \frac{4}{5}(18.75) + \frac{4}{5}(-43.75) = 0$$

Check these results by starting from joint A!

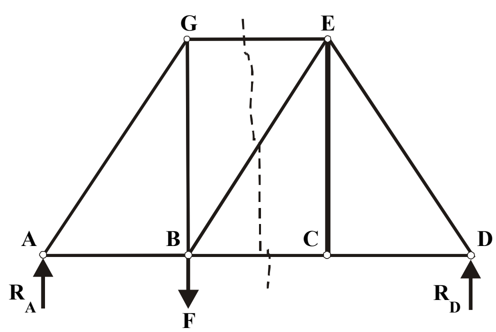
Method of sections

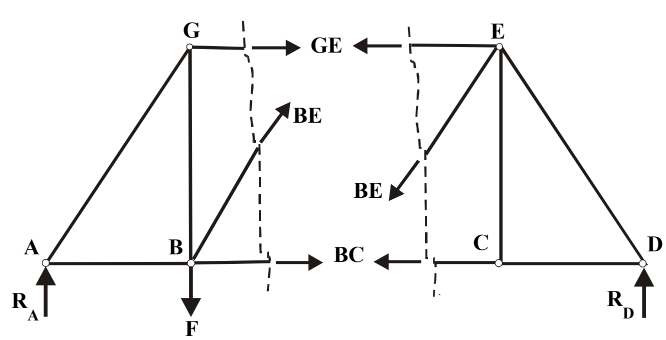
In the previous method of joints we used only two of the three equations of equilibrium. We can take advantage of the three equations by sectioning a frame and by selecting one section for a free body diagram. External forces acting in the member's direction replace the 'cut' members in the section. The main advantage of this method is that internal forces can be found directly without the need to carry out joint to joint calculations. Using the method of sections the following rules should be followed:

- No more than 3 members to be cut.
- The external reaction forces are not to be removed.
- Section internal member forces are not considered in the FBD analysis of the section.

In some cases the methods of sections and joints can be combined for efficient solution.

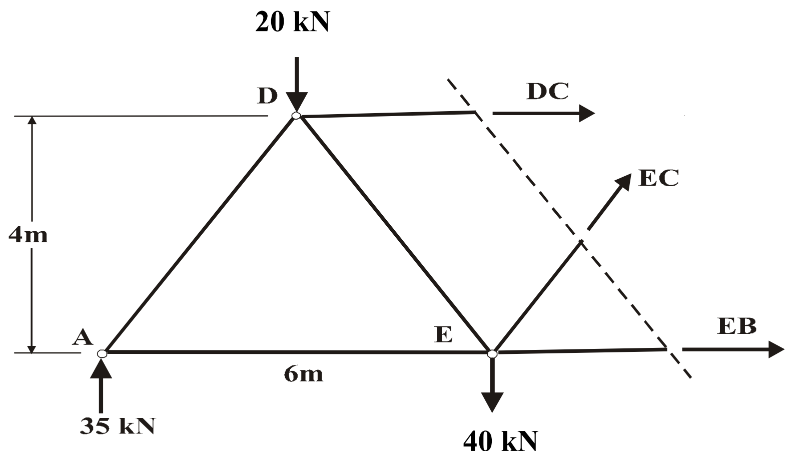
The method of sections is illustrated by using the frame below. The frame is cut through members GE, BE and BC. For each of the two sections we have three unknown external forces (the reactions are calculated initially for the FBD of the entire frame) and three equations: the sum of the forces in the horizontal and vertical is zero and the sum of moments at any point is zero.





Example 2 (method of sections)

Revisiting the frame used for the joints method sample problem, we may section this frame at members DC, EC and EB and resolve directly the member forces as shown below:



$\Sigma F_y = 0$

$$\frac{4}{5}EC + 35 - 20 - 40 = 0$$

$$EC = 31.25kN \quad (\text{tension})$$

$\Sigma M_d = 0$ (+)

$$3(35) + 3(40) - 4(EB) - 4\left(\frac{3}{5}EC\right) - 3\left(\frac{4}{5}EC\right) = 0$$

$$EB = 18.75kN \quad (\text{tension})$$

$\Sigma F_x = 0$

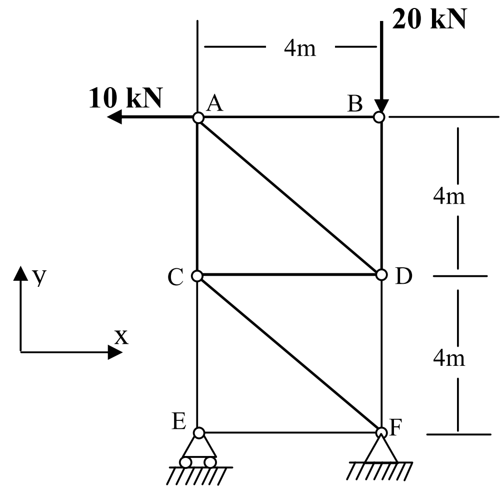
$$DC + EB + \frac{3}{5}EC = 0$$

$$DC = -37.5kN \quad (\text{compression})$$

Note the agreement with the forces resolved using the method of joints.

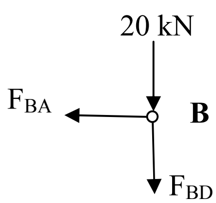
Example 3

Determine the forces in members AC, AD and BD for the frame shown.



Method of joints:

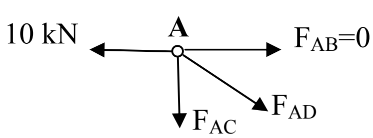
<u>Joint B</u>



$\Sigma Fx = 0 \quad -F_{BA} = 0 \quad F_{BA} = 0$

$\Sigma Fy = 0 \quad -20-F_{BD} = 0 \quad F_{BD} = -20$ kN

<u>Joint A</u>



$\Sigma Fx = 0 \quad -10+F_{AD}\mathrm{Cos}\ 45^0 = 0 \quad F_{AD} = 14.142$ kN

$\Sigma Fy = 0 \quad -F_{AC} - F_{AD}\mathrm{Sin}45^0 = 0 \quad F_{AC} = -10$ kN

Method of sections:

Section through AC, AD and BD and draw a FBD.

Summary

Two methods often used to calculate member's internal forces in frames under statics load.

Method of joints

- Usually used to find ALL the forces in the members of a frame.
- At any joint in a plane frame no more than two are unknown.
- Vertical and horizontal force equations are solved for each joint simultaneously.
- Proceeding systematically through the frame.
- Usually reaction forces are determined first.

Method of sections

- Used to find internal forces in 3 particular members of a frame.
- The frame is sectioned where no more than 3 members are cut.
- A FBD of one section is drawn with forces in the cut members at the direction of the members.
- The three plane equilibrium equations are used to calculate all forces.

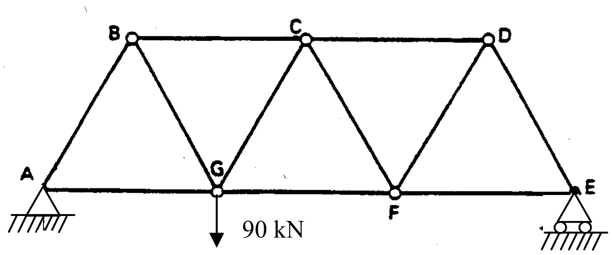
Further readings:

Space structures

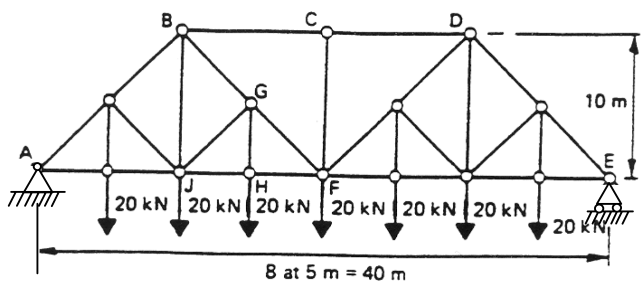
Other methods exists such as the **method of coefficients**.

Tutorials

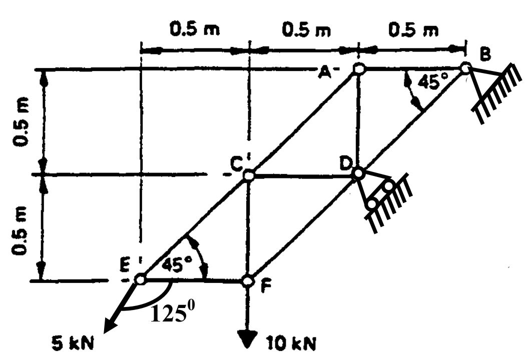
1. The truss below is constructed with similar members all of which are 3m long. Determine the forces in all the members due to the external vertical load of 90 kN at G.



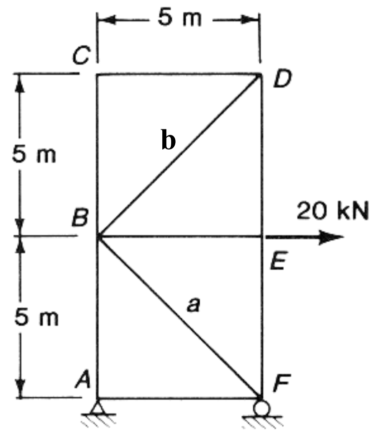
2. Determine the forces in members GF and HF for the truss shown. Hence, determine the forces of the members meeting at joint G.

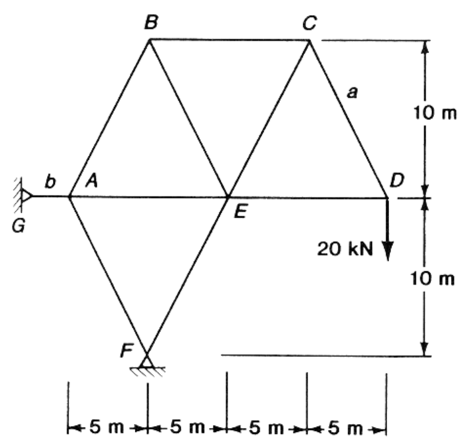


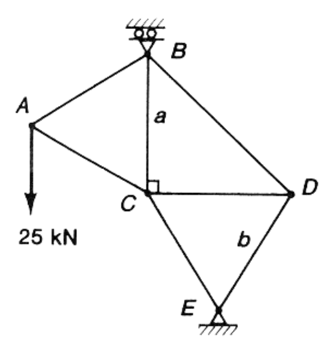
3. Determine why the frame shown is statically determinate and find the forces in the members when the loads are applied.



4. Calculate the member forces in the letter members (a and b) of the trusses shown below.







(All members are 5m long except BD)

Shear Force and Bending Moment analysis of Beams

Introduction

Beams are important structural and mechanical engineering elements. In this lecture we will determine the internal loading of planar beams caused by bending.
To calculate the largest internal shear force and bending moment which develop in a beam the shear force and bending moment diagrams are drawn using statics principles.

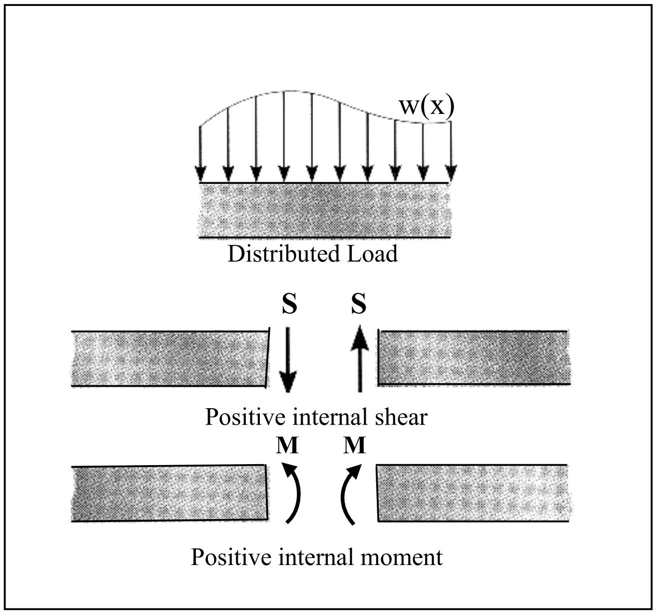
In our present discussion we are restricted to beams that carry loads perpendicular to their longitudinal axes and have a symmetric cross section. The beams are assumed to be straight and made of homogeneous material. The material behaves elastically and is below the limit of proportionality.

First step: Beam Free Body Diagram

Prior to the SF and BM analysis we must construct the beam as a FBD and resolve the beam reaction forces using the statics methods shown previously.

Beam Sign convention

Before plotting the shear force and bending moment diagram it is first necessary to establish a sign convention so as to define the positive and negative numerical values. We use the convention that is often used in engineering where internal shear force causes a ***clockwise*** *rotation* to the beam section investigated and the bending moment causes ***compression in the top of the beam***. This sign convention is illustrated below. The positive moment sign causes such bending in the beam that the top can 'hold water'.



Beam internal shear force and bending moment.

From beam's deformation theory it is possible to show that the slope of a beam shear diagram is equal to the intensity of the beam resultant external load at every point, with a negative sign:

$$\frac{dS}{dx} = -w(x)$$

[slope of shear diagram] = −[distributed load]

Similarly it is also possible to demonstrate that the slope of the moment at every beam cross section is equal to the shear value:

$$\frac{dM}{dx} = S$$

[slope of the moment diagram] = [shear]

If we integrate the force along the beam length between any two points we can obtain the negative change in shear:

$$\boxed{\begin{array}{c} \Delta S = -\int w(x)dx \\ \text{[change in shear]} = -\text{[area under distributed load]} \end{array}}$$

and, similarly, if we integrate the shear force along a particular beam section we can obtain the change in the bending moment:

$$\boxed{\begin{array}{c} \Delta M = \int S(x)dx \\ \text{[change in moment]} = \text{[area under shear diagram]} \end{array}}$$

In this lecture we will establish simple rules for drawing the SF and BM diagrams that are the direct results of these relationships.

General rules for drawing the shear force (SF) and bending moment (BM) diagrams:

1. Concentrated point force:
Will change the **SF** diagram by the value of the load at that point.
Will change the **BM** diagram slope from the initial to the new shear force.

2. Point moment (a couple):
No change in the **SF** diagram.
Will change the **BM** diagram by the value of the moment at that point.

3. Uniformly distributed load (UDL):
A constant negative **straight** slope of the **SF** diagram with the total change equal to the total load applied.

A positive slope of the **BM** diagram with total change equal to the change in the bending moment.

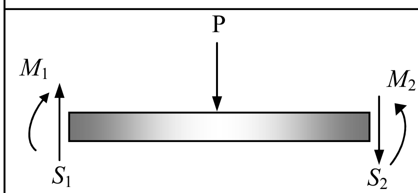
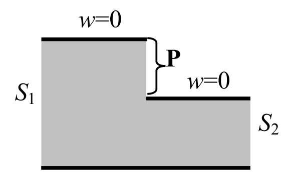
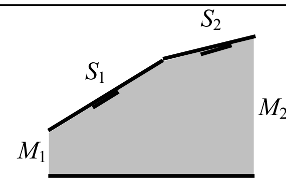
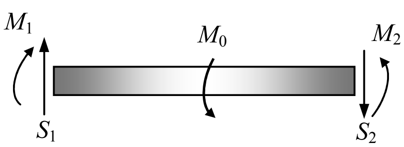
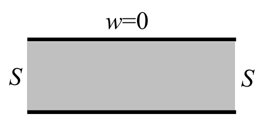
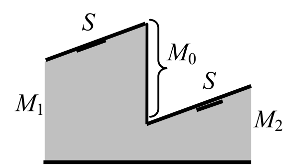
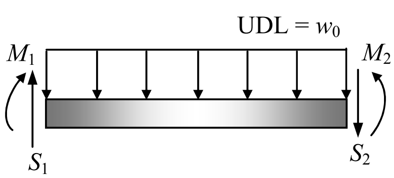
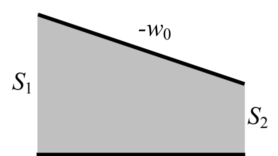
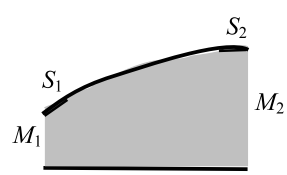
4. Increased distributed load:
A negative slope of the **SF** diagram with the total change equal to the total applied load.
A positive slope of the **BM** diagram with total change equal to the change in the bending moment.

5. Decreased distributed load:
A negative **concave** slope of the **SF** diagram with the total change equal to the total applied load.
A positive slope of the **BM** diagram with total change equal to the change in the bending moment.

Special points to calculate the BM diagram

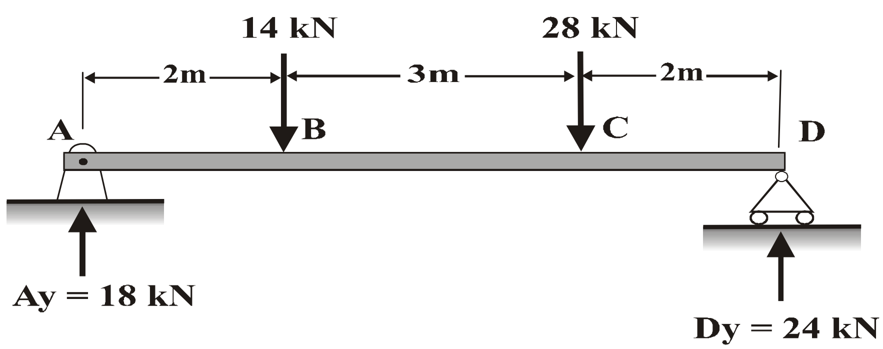
- Where the **shear force is zero** – **BM moment is maximum**.
  This is important for the calculation of the maximum bending stress.
- Where point loads are applied.
- Where point couples are applied.
- At the start and end of an applied UDL.

Variation of the Shear Force and Bending Moment for Common loading of beams:

Loading	Shear Diagram $\frac{dS}{dx} = -w$	Moment Diagram $\frac{dM}{dx} = S$
	 Downward force **P** causes S to jump downward from S_1 to S_2.	 Constant slope changes from S_1 to S_2.
	 No change in shear since slope $w = 0$.	 Constant positive slope. Counterclockwise M_0 causes M to jump downward.
	 Constant negative	 Positive slope that decreases from S_1 to S_2.

Example 1

For the simply supported beam shown below derive the SF and BM equations for each segment of the beam and sketch the SF and BM diagrams. Ignore the weight of the beam. Note that the reaction forces have already been computed.



Solution:

<u>Segment AB (0 < x < 2m):</u>

$$\Sigma F_y = 0 \qquad 18 - S = 0$$
$$S = 18\text{kN}$$
$$\Sigma M = 0 \qquad -18x + M = 0$$
$$M = 18x \ \text{kN m}$$

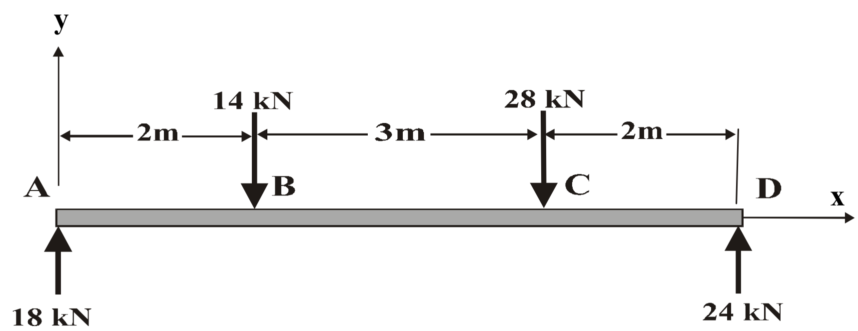
<u>Segment BC (2m < x < 5m):</u>

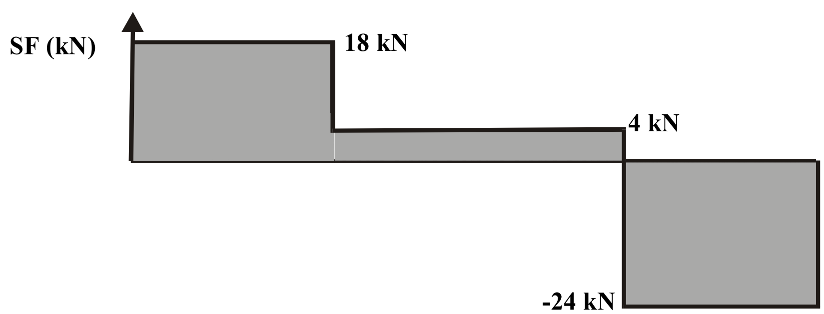
$$\Sigma F_y = 0 \qquad 18 - 14 - S = 0$$
$$S = 4 \ \text{kN}$$
$$\Sigma M = 0 \qquad -18x + 14(x-2) + M = 0$$
$$M = 4x + 28 \ \ \text{kN m}$$

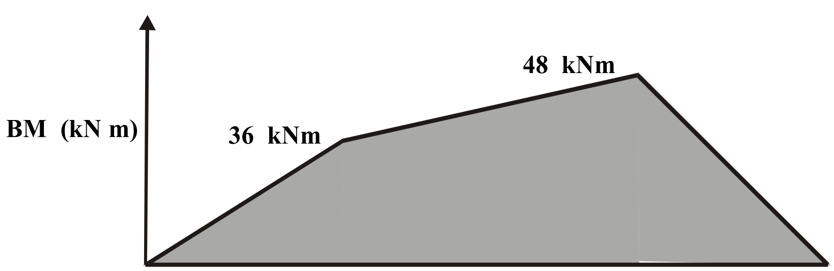
<u>Segment CD (5m < x < 7m):</u>

$$\Sigma F_y = 0 \qquad 18 - 14 - 28 - S = 0$$
$$S = -24 \ \text{kN}$$
$$\Sigma M = 0 \qquad -18x + 14(x-2) + 28(x-5) + M = 0$$
$$M = -24x + 168 \ \ \text{kN m}$$

SF and BM diagrams:







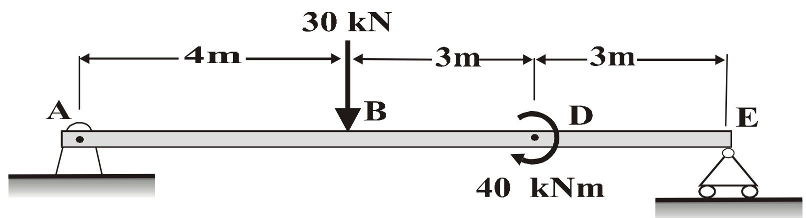
Comments:

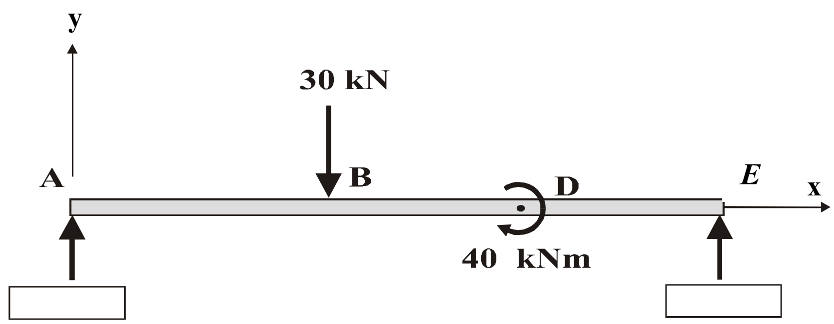
- In this simple case it is only necessary to calculate the SF and BM at points B, C.
- The maximum BM is at C because the SF=0.
- In the general case the exact location of the maximum BM can be found by equating the SF force equation to zero.
- Note that the point moments are equal to the corresponding areas under the SF diagram.

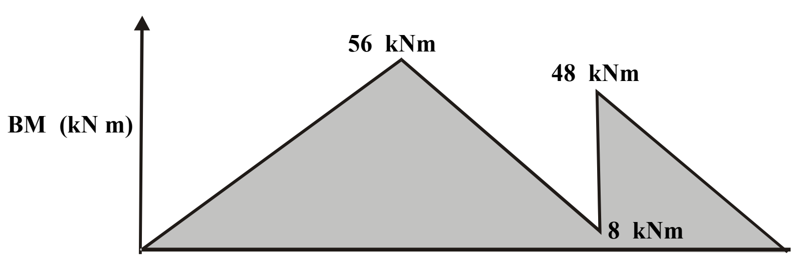
Example 2

For the following beam under a concentrated load of 30kN and a couple of 40kNm the BM diagram is shown. Calculate:
Reaction forces.
Shear force
Draw the shear force diagram.

Calculations:

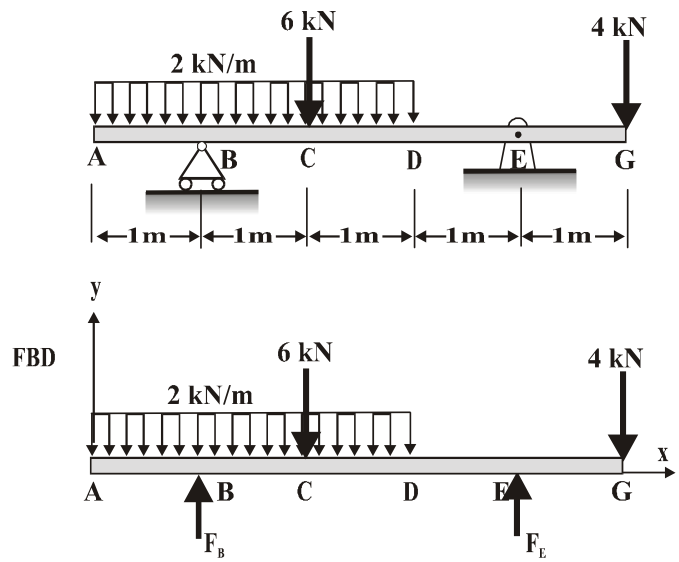


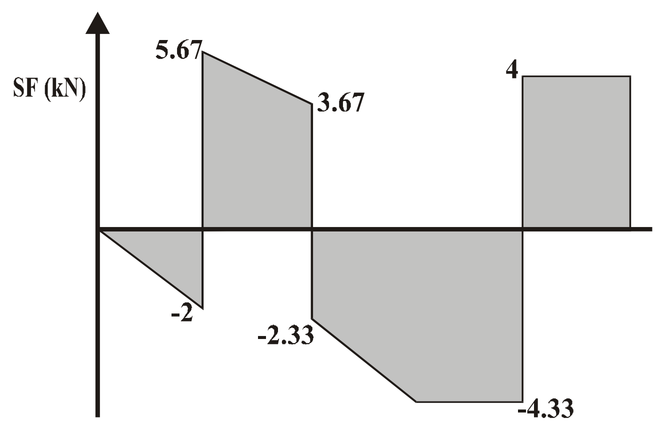


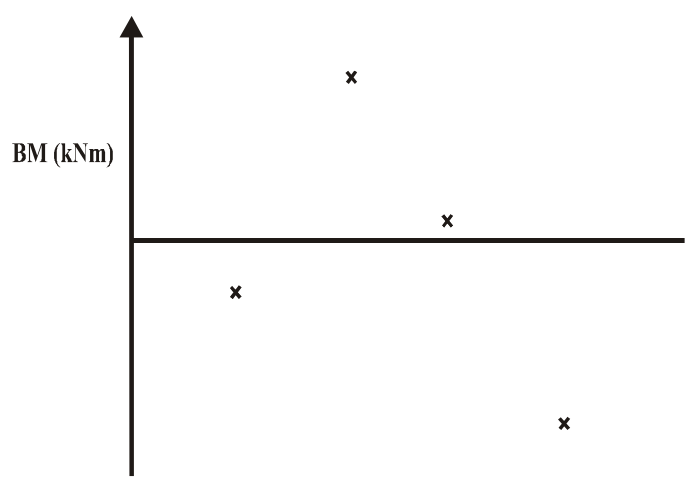


Example 3

For the beam shown complete the following:
The reaction forces at B and E.
Compute the values of the BM at B, C, D and E.
Plot the BM diagram.

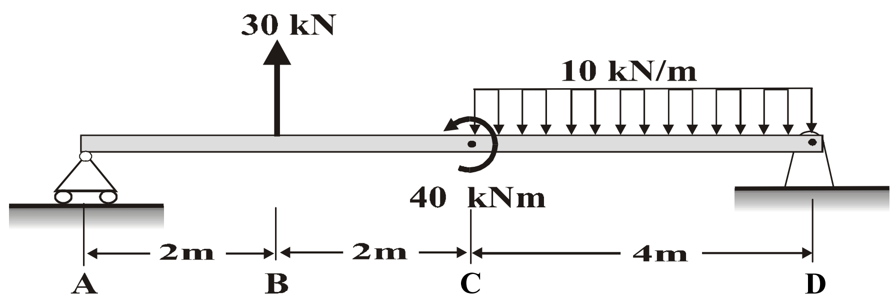






Tutorials

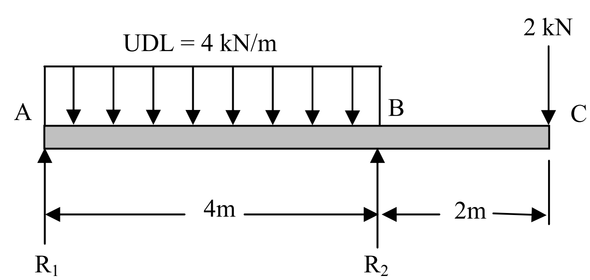
1. Plot the shear force and bending moment diagrams for the following beam and compute the values at all important locations.



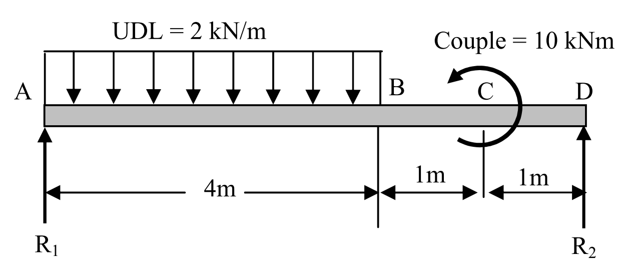
(solution to this problem at the end of the section)

Determine the reactions, draw the SF and BM diagrams, and determine the maximum BM and their positions for the following beams:

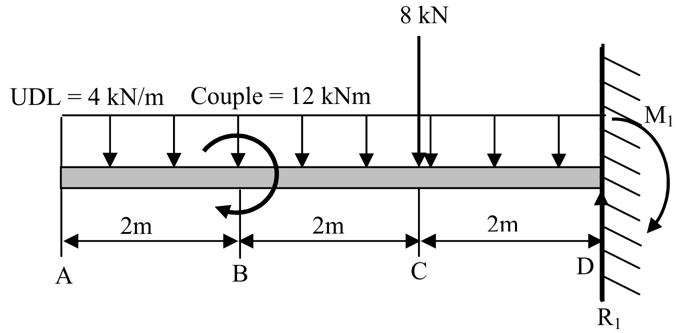
2.



3.

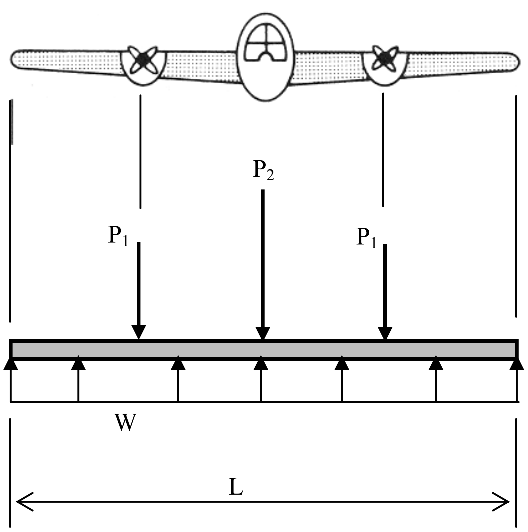


4.

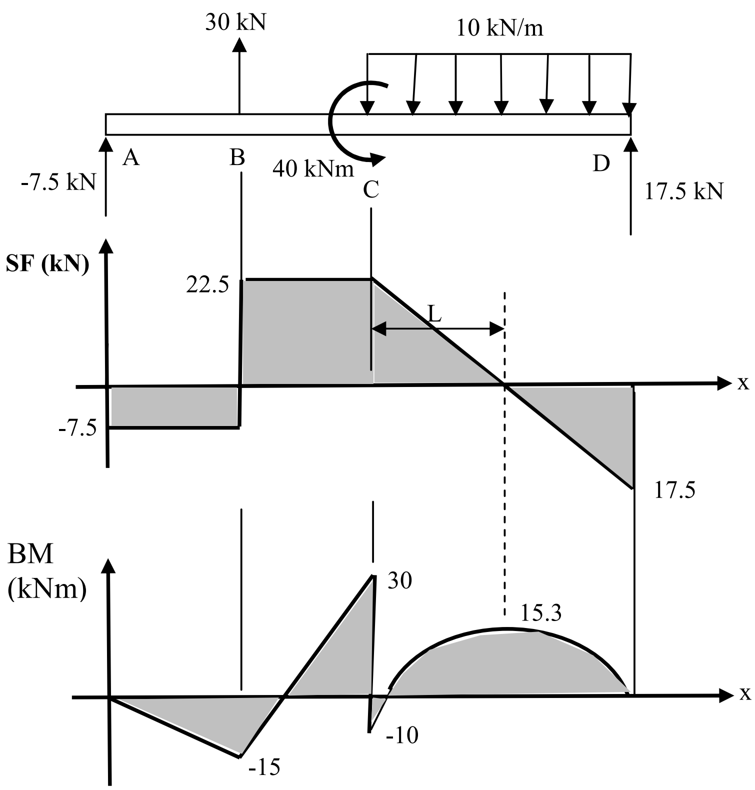


5. The figure below shows a twin engine plane where the weight of the two engines is denoted by P_1, and P_2 is the gross weight of the fuselage. To maintain equilibrium that keeps the plane in the air, the wings generate an uplift force represented by the uniformly distributed load W. (assume P2 = 1.25P1)

a. Construct a shear force diagram.
b. Determine the maximum shear force resisted by the wings.
c. Construct a bending moment diagram.
d. Calculate the position and value of the maximum bending moment.



Solution to tutorial No. 1:



Part solutions:

Tutorial No.	R_1 kN	R_2 kN	Max SF kN	Position m	Max BM kNm	Position m
2	7	11	-9	4	6.125	1.75
3	7	1	7	0	12.25	3.5
4	32	M_1=76	-32	6	-76	6

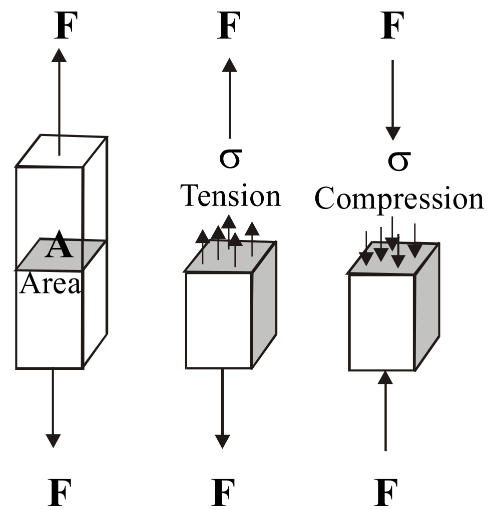
Stress and Strain

Normal (direct) Stress

The intensity of force, or force per unit area, acting normal to an area is defined as the normal stress, σ (sigma).

Providing that a cross section of a component is under a constant normal force, the normal stress is the force divided by the area:

$$\boxed{\text{stress } \sigma = \frac{\text{Force}}{\text{Area}} = \frac{F}{A}}$$



σ = Average normal stress at any point on the cross section.
A = Cross section area.
F = Internal resultant force normal to the area.

If the normal force F 'pulls' the area A, it is a **tensile stress**, whereas if it 'pushes' the area it is a **compressive stress**.

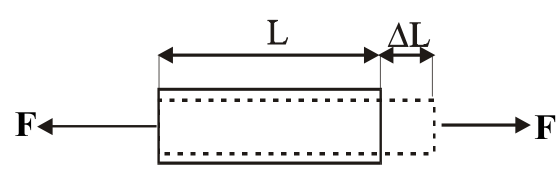
Units – the SI units used are Newtons per square meter [N/m^2]. This is called 1 Pascal and is a small unit. It is customary in engineering to use MPa as follows:

$$\boxed{10^6\ N/m^2 = 1\ MN/m^2 = 1\ N/mm^2 = 1\ MPa}$$

Normal (Direct) strain

The change in length of a member due to the internal normal load is called normal strain. In other words, the relative deformation of a material per unit length is defined as the normal strain ε (epsilon):

$$\boxed{\text{strain } \varepsilon = \frac{\text{Change of length}}{\text{Original length}} = \frac{\Delta L}{L}}$$



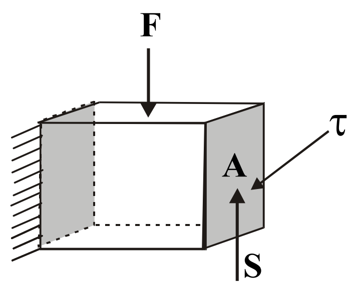
Units – Strain is a dimensionless quantity, in SI units: [mm/mm].

Shear Stress

The intensity of force, or the force per unit area, acting **tangent** to an area is called the shear stress, τ (tau).

The shear stress in a beam section is the shear force divided by the section area.

$$\boxed{\text{Shear stress } \tau = \frac{\text{Shear Force}}{\text{Area}} = \frac{S}{A}}$$

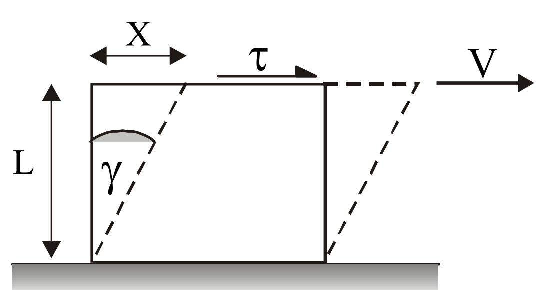


Shear strain

The shear strain is the change in angle due to distortion that occurs between two faces that were originally perpendicular to one another. It is denoted by γ (gamma) and measured in radians (rad).

We assume that the distortion angle is small and therefore: $\tan(\gamma) = \gamma$. This makes the shear strain's non-dimensional quantity similar to the normal strain.

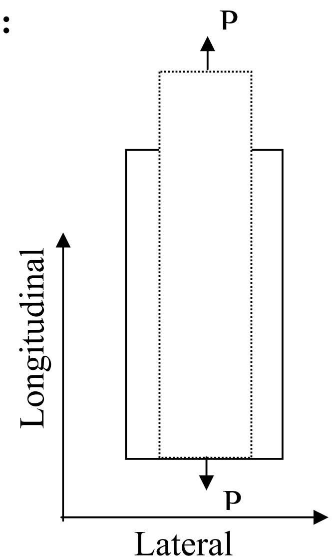
$$\boxed{\text{Shear strain } \gamma(\text{rad}) = \tan(\gamma) = \frac{X}{L}}$$



*Assume small strains.

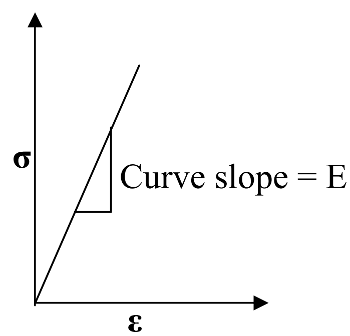
Elastic constants

Poisson's Ratio:



$$\boxed{\upsilon = -\frac{\text{Lateral strain}}{\text{Longitudinal strain}} = -\frac{\varepsilon_x}{\varepsilon_y}}$$

Young Modulus (Elastic Modulus):



$$\boxed{E = \frac{\text{Stress}}{\text{strain}} = \frac{\sigma}{\varepsilon}}$$

Shear Modulus:

$$\boxed{G = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{\tau}{\gamma}}$$

and also:

$$\boxed{E = 2G(1+\nu)}$$

Units: υ - no units,
G and E - $[GN/m^2] = [GPa]$

Allowable stress

A design engineer of a structure or machine must restrict the stress in the material of all the designed elements to a level that is safe under the operational conditions. To ensure the safety, an **allowable stress** is chosen that restricts the maximum loading.

One method of specifying an **allowable stress** is to use a **factor of safety** (FOS). That is the ratio between the maximum stress or load at failure and the allowed stress or load:

$$\boxed{FOS = \frac{\sigma_{failed}}{\sigma_{allowed}}}$$

The FOS is dependent on each specific application. A typical value that is near 1.5 may be used for the aerospace industry in order to save weight, and a FOS value closer to 3 for the design of nuclear power plant.

In statics design it is common to assume that the maximum stress before failure is equal to the material yield stress, σ_y. Therefore, to obtain the maximum allowed stress in the design, $\sigma_{allowed}$, for a given material:

$$\boxed{\sigma_{allowed} = \frac{\sigma_y}{FOS}}$$

Example 1

A tie bar in a roof truss is 20mm diam. and 4m long. It carries a tensile load of 80 kN. If E=200GN/m^2, calculate the elongation under this load.

Solution:

Cross section area: $A = \pi D^2/4 = \pi\ 20^2/4 = 314.16\ mm^2$

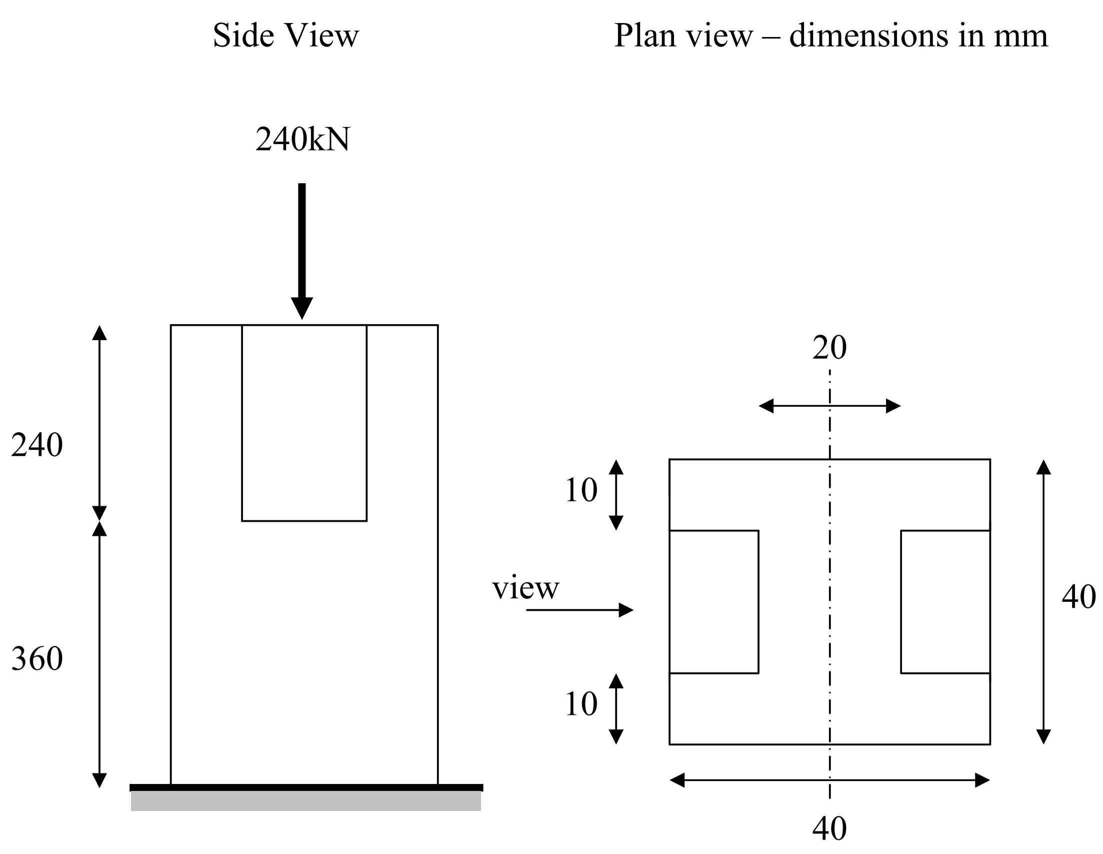
Axial stress: $\sigma = F/A = 80000/314.16 = 254.65$ MPa

Axial strain: $\varepsilon = \sigma/E = 254.65/200000 = 0.0012732$

From definition: strain $\varepsilon = \dfrac{\Delta L}{L}$ → Elongation $\Delta L = L\varepsilon = 5.09$mm

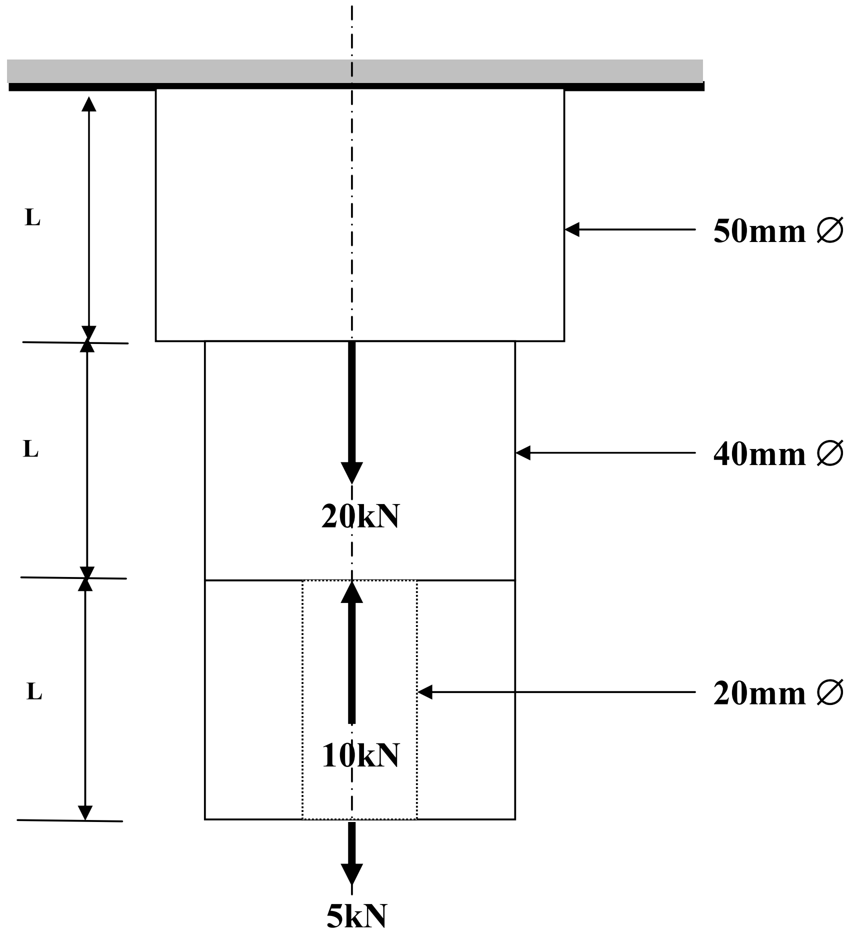
Example 2

A steel strut has two grooves cut along part of its length. Calculate the total compression of the strut due to a load of 240kN. Young's Modulus = 200GN/m^2.



Example 3

A steel rod of varying cross-section is loaded as shown below. Determine where the maximum stress occurs and the maximum deflection. (E = 210 GPa, L = 50mm).



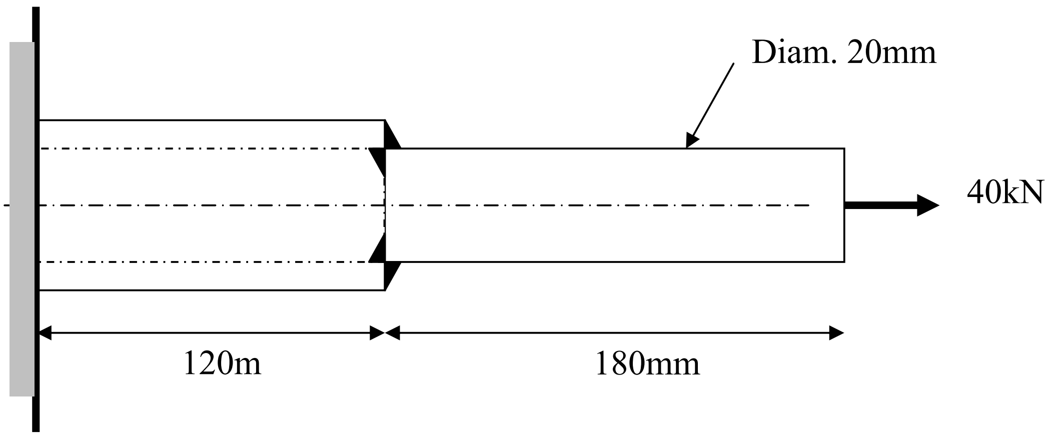
Example 4

A solid cylindrical bar, of 20mm diameter and 180mm long, is welded to a hollow tube of 20mm internal diameter, 120mm long, to make a bar of total length 300mm.

Determine the external diameter of the tube if, when loaded axially by a 40kN load, the stress in the solid bar and that in the tube are to be the same.

Hence calculate the total change in length of the bar and the maximum deflection.

$E = 210\ kN/mm^2$ (or $210\ GN/m^2$)



Solution Example 2

For solid length: $\sigma = \dfrac{240 \times 10^{3}}{0.04 \times 0.04} = 0.15 \times 10^{9}\ N/m^{2}$

Strain, $\varepsilon = \dfrac{\sigma}{E} = \dfrac{0.15 \times 10^{9}}{200 \times 10^{9}} = 0.00075$

For Grooved length: $\sigma = \dfrac{240 \times 10^{3}}{(0.04 \times 0.04) - (0.02 \times 0.02)} = 0.2 \times 10^{9}\ N/m^{2}$

Strain, $\varepsilon = \dfrac{\sigma}{E} = \dfrac{0.2 \times 10^{9}}{200 \times 10^{9}} = 0.001$

Total strut compression = $\Delta L_S + \Delta L_G$ = 360×0.00075 + 240×0.001 = 0.51mm

Solution Example 3

Net force in top section = 20+5- 10 = 15kN $\sigma = \dfrac{15\times10^{3}}{\left(\dfrac{\pi}{4}\right)(0.05)^{2}} = 7.64MN/m^{2}(+ve)$

Net force in mid section = - 10 +5 = -5kN $\sigma = \dfrac{-5\times10^{3}}{\left(\dfrac{\pi}{4}\right)(0.04)^{2}} = -3.98MN/m^{2}(-ve)$

Net force in bottom section = 5kN $\sigma = \dfrac{5\times10^{3}}{\left(\dfrac{\pi}{4}\right)\left(0.04^{2}-0.02^{2}\right)} = 5.3MN/m^{2}(+ve)$

Solution Example 4

Stress in solid bar = F/A = $\dfrac{40\times10^{3}}{(\pi/4)\left(0.02^{2}\right)} = 127.3MN/m^{2}$

Stress in hollow tube = F/A = $\dfrac{40\times10^{3}}{(\pi/4)\left(d^{2}-0.02^{2}\right)} = \dfrac{50929.58}{\left(d^{2}-0.02^{2}\right)}$

Equating these stresses: $127.3\times10^{6} = \dfrac{50929.58}{\left(d^{2}-0.02^{2}\right)}$

Therefore: d = 0.02828m or 28.3mm

Strain for whole bar = $\dfrac{127.3\times10^{6}}{210\times10^{9}} = 6.062\times10^{-4}$

Hence, $\Delta L = 6.062\times10^{-4} \times 300 = 0.182mm$

Tutorials

1. Part of an aircraft undercarriage consists of a hollow alloy tube of 75mm outside diameter. The maximum applied axial load is 80 kN and the maximum material stress is 62 MPa. If the tube is to be as light as possible determine the minimum thickness of the tube wall. Assume a factor of safety of 2 and elastic modulus E=115 GPa.

Solution procedure:

- Find the allowable stress.
- Use the direct stress definition.
- Rearrange to obtain the internal diameter.
- Determine the thickness.

2. A steel column is 900mm long and 150mm diam. If a hole 100mm diam. and 600mm long is bored out, calculate the total compression of the column when carrying a load of 1 MN and E = 207 GPa.

3. A tensile test was carried out on a wire of 0.38mm diam. The extension of the wire was measured over a length of 4.6m. A graph of the load against extension was plotted and a gradient of 4.9 N/mm was obtained. Calculate the wire Modulus of elasticity.

Solutions: 1. 13.32mm, 2. 0.377mm, 3. 198.75 GPa

Stresses in Beams

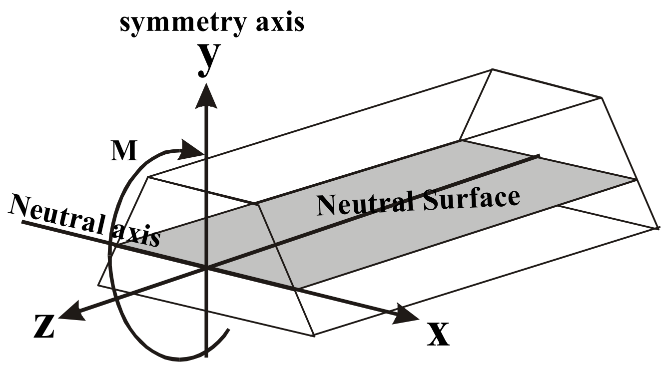
Introduction

In this section we develop and learn how to use the engineering bending equation often called the **flexural formula**. It is one of the most important relationships in beam theory and is frequently used by engineers to approximate the strength of beam structures.

The engineering bending equation allows direct stress calculation from the applied moment and beam geometry when certain conditions are maintained.

Prior to the deformation we assume:

- A straight prismatic body.
- Homogeneous and isotropic material.
- Symmetrical cross section.
- An applied moment perpendicular to the axes of beam symmetry.



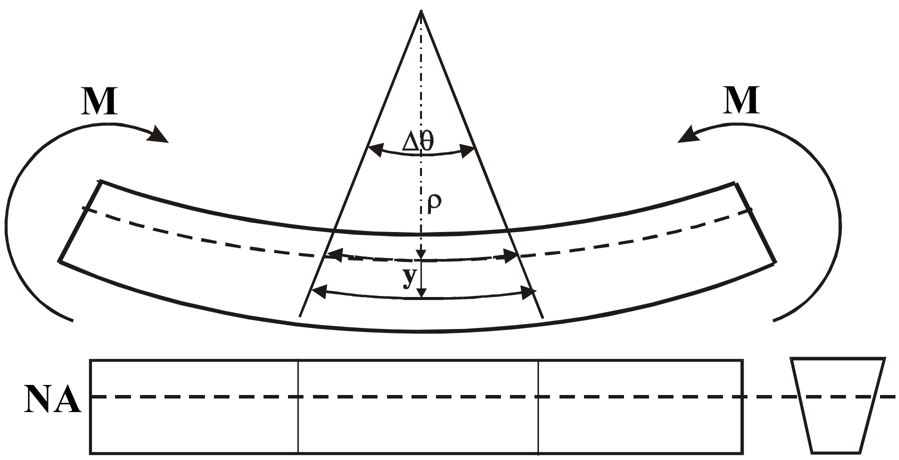
- The intersection of the **Neutral surface** with any cross-section plane is called the **Neutral axis (NA).**

After the deformation we assume:

- The longitudinal axes do not change in length but become a curve.
- All the beam cross sections remain plane and perpendicular to the longitudinal axes.
- The deformation of a cross section within its plane is negligible.

1. Normal strain in beams

Assume a beam section is under a moment as shown below.
The beam deflects with a radius of curvature equal to ρ:



The beam fibres in the **convex, lower part are elongated** and those on the **concave, upper part are shortened**. Thus, the fibre in the lower part of the beam is in **TENSION**, whereas those in the upper side are in **COMPRESSION**.

Somewhere in between these two extremes an axis with no deformation at all exists. We call this axis the **Neutral Axis (NA)**.

Along the NA we assume that no change in length has occurred, therefore from geometrical considerations:

$$\boxed{\Delta x = \rho\,\Delta\theta}$$

If we consider a fibre at a typical distance y from the neutral axes, the length of this fibre is:

$$\boxed{\Delta x' = \Delta\theta(\rho + y)}$$

Using the strain definition, the longitudinal strain is equal to the elongation divided by the initial length:

$$\boxed{\varepsilon_x = \frac{(\Delta x' - \Delta x)}{\Delta x} = \frac{(\rho + y)\Delta\theta - \rho\,\Delta\theta}{\rho\,\Delta\theta} = \frac{y}{\rho}}$$

Hence:

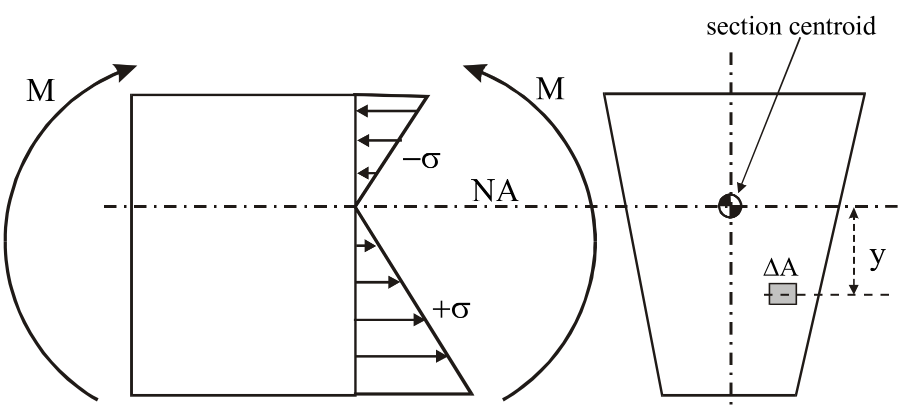
$$\varepsilon_x = \frac{y}{\rho} \qquad \text{For fibers under tension}$$

$$\varepsilon_x = -\frac{y}{\rho} \qquad \text{For fibers under compression}$$

Since stress is uniaxial: $\boxed{\varepsilon_x = \frac{\sigma_x}{E}}$ and finally: $\boxed{\frac{\sigma_x}{E} = \frac{y}{\rho}}$

2. Location of the Natural Axis (NA)

Inspecting a section of a beam with arbitrary cross section and applied moment as shown, we consider an element of area ΔA at a distance y from the NA.



The normal force acting on this element has a magnitude: $\boxed{\sigma \Delta A}$

The total force is the sum, or the integration, of the force acting on the cross section and, to satisfy static equilibrium, it must be equal to zero:

$$\boxed{\int_A \sigma \Delta A = 0}$$

Substituting with the elastic modulus definition: $\boxed{\sigma = E\varepsilon_x}$ and $\boxed{\varepsilon_x = \frac{y}{\rho}}$ we obtain:

$$\boxed{\int_A \left(\frac{Ey}{\rho}\right)\Delta A = 0}$$

However, since the radius of curvature, ρ, and the modulus of elasticity, E, cannot be zero,

$$\boxed{\frac{E}{\rho} \neq 0}$$

we conclude that: $$\boxed{\int_A y\Delta A = 0}$$

The meaning of this equation is that **THE FIRST MOMENT OF AREA** of the beam is zero.

And hence:
" ***The neutral axis passes through the centroid of the beam cross section"***. We use this property to calculate the LOCATION of the Neutral axis with respect to the y-axis.

The y-axis LOCATION of the Neutral Axis for the TOTAL cross section is defined as:

$$\boxed{\bar{y} = \frac{\int_A y\,\Delta A}{\int_A \Delta A} = \frac{\sum (y_i\,\Delta A_i)}{\sum (\Delta A_i)}}$$

Where y_i and A_i correspond to a particular cross section i of the beam. (more on this later).

3. The normal stress equation (the flexural formula)

For a beam under static load at any cross section, the applied moment must be equal to the internal moment.

The applied moment from a small area, dA, about the NA is:

$$\boxed{dM = y\,dF = y(\sigma_x\,dA)}$$

And the total internal moment is the sum of the moments in the cross section:

$$\boxed{M_{cross\,section} = \int y\,\sigma_x\,dA = \int y\left(\frac{Ey}{\rho}\right)dA}$$

Since the total sum of the moments is zero, this must be equal to the applied moment. Also, E and ρ are constants:

$$\boxed{M_{applied} = \int y\left(\frac{Ey}{\rho}\right)dA = \frac{E}{\rho}\int y^2 dA}$$

The expression: $\boxed{\int y^2 dA}$ in the above equation is an important geometry quantity called the:

SECOND MOMENT OF AREA ABOUT ITS NA – I_{NA}

Where: $\boxed{I_{NA} = \sum y^2 dA}$

Substituting from the previous relation: $\boxed{\dfrac{E}{\rho} = \dfrac{\sigma}{y}}$

And rearranging we eventually arrive at the equation:

Or, the

FLEXURAL FORMULA:

$$\boxed{\sigma = \frac{M\bar{y}}{I}}$$

Where:

M =Applied moment at the beam cross section considered

I = Second moment of area of the beam around the NA.

$\bar{y}$ = Distance from the NA.

The maximum stresses will be at y_{max} on the beam top or bottom faces.

And finally, for a simple bending of a beam the complete relationship between the beam deflection, moment and the section stress is:

$$\boxed{\frac{M}{I} = \frac{E}{\rho} = \frac{\sigma}{y}}$$

Section Modulus

A widely used geometrical property of a beam section is the ratio between the second moment of area, I, to the maximum distance from the NA, y_{max}, and is called the Section Modulus, Z (not to be confused with Elastic Modulus, E).

$$\boxed{Z = \frac{I}{y_{max}}}$$

Where Z_{xx} refers to a moment applied about the beam x-axis and Z_{yy} is for a moment applied about the beam y-axis.

The values of section modulus of standard beams are tabulated in engineering data handbooks.

Using the section Modulus, Z, in the engineering bending equation above, we obtain the maximum stress in a beam section due to bending:

$$\boxed{\sigma = \frac{M}{Z}}$$

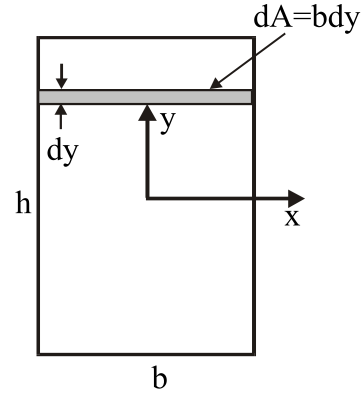
4. Calculation of the Second Moment of Area (I)

The calculation of the second moment of (plane) area is carried out with respect to the section x-axis or y-axis:

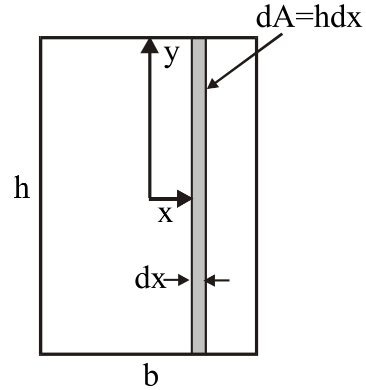
$$\boxed{I_{xx} = \int_A y^2 dA \quad \text{and} \quad I_{yy} = \int_A x^2 dA}$$

This can be demonstrated using a rectangular cross section area:

$$I_{xx} = \int_A y^2 dA = \int_{-h/2}^{h/2} y^2 b\,dy = \left.\frac{b y^3}{3}\right|_{-h/2}^{h/2} = \frac{bh^3}{12}$$



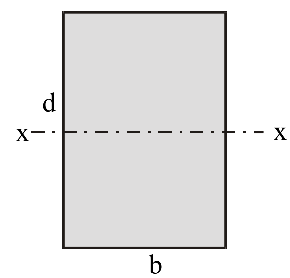
$$I_{yy} = \int_A x^2 dA = \int_{-b/2}^{b/2} x^2 h dx = \left.\frac{h x^3}{3}\right|_{-b/2}^{b/2}$$
$$= \frac{hb^3}{12}$$



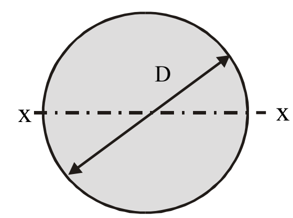
Note that for rectangular section $I_{yy} < I_{xx}$. How does this affect the beam strength?.

5. Second moment of area of simple cross sections:

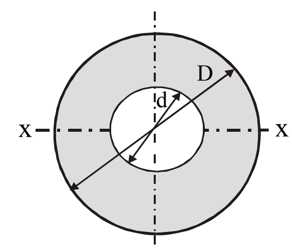
Rectangular section: $I_{xx} = \frac{bd^3}{12}$



Solid circular section $I_{xx} = \frac{\pi D^4}{64}$

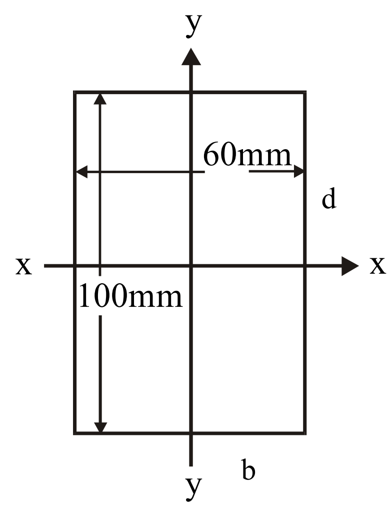


Hollow circular section: $I_{xx} = \frac{\pi\left(D^4 - d^4\right)}{64}$



Example 1

A beam with the following rectangular cross section is under a maximum bending moment of 5 kNm. Find the maximum stress in the beam if this moment is applied 1. About the x-axis and, 2. About the y-axis.



1. About the x-axis:

$$I_{xx} = \frac{bd^3}{12} = \frac{60x\,100^3}{12} = 5x\,10^6 \quad mm^4$$

$$Z_{xx} = \frac{I_{xx}}{y_{max}} = \frac{5x10^6}{50} = 10x10^4 \quad mm^3$$

$$\sigma_{xx} = \frac{M}{Z_{xx}} = \frac{5000x\,10^3}{10x\,10^4} = 50 \quad MPa$$

2. About the y-axis:

$$I_{yy} = \frac{db^3}{12} = \frac{100x\,60^3}{12} = 1.8x\,10^6 \quad mm^4$$

$$Z_{yy} = \frac{I_{yy}}{y_{max}} = \frac{1.8x10^6}{30} = 6x10^4 \quad mm^3$$

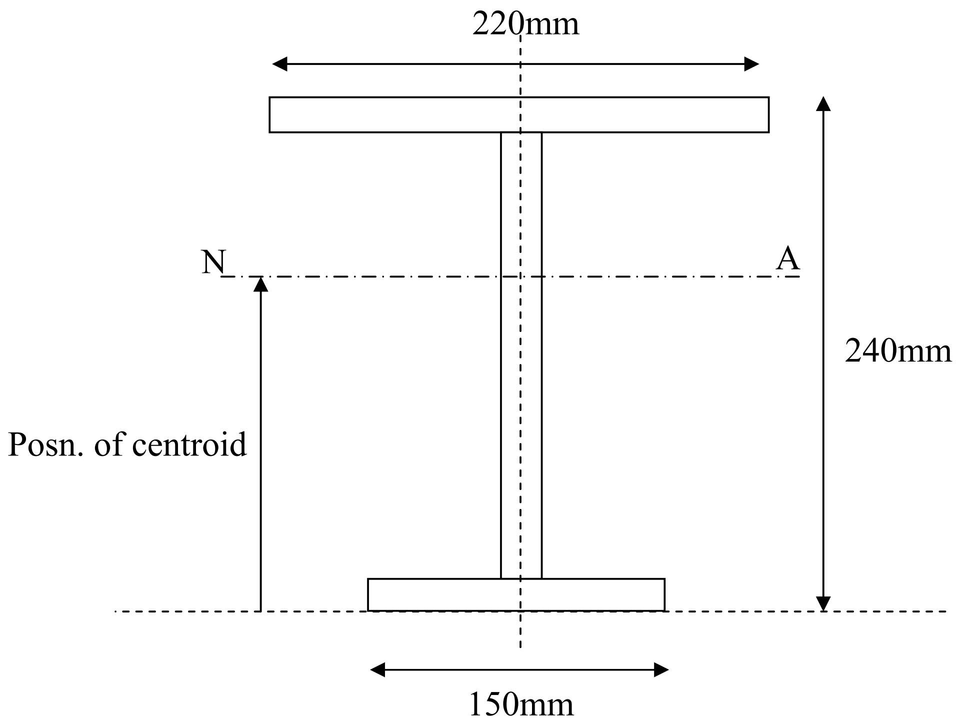
$$\sigma_{yy} = \frac{M}{Z_{yy}} = \frac{5000x\,10^3}{6x\,10^4} = 83.33 \quad MPa$$

What are the implications for the beam strength?

Additional Example to solve later

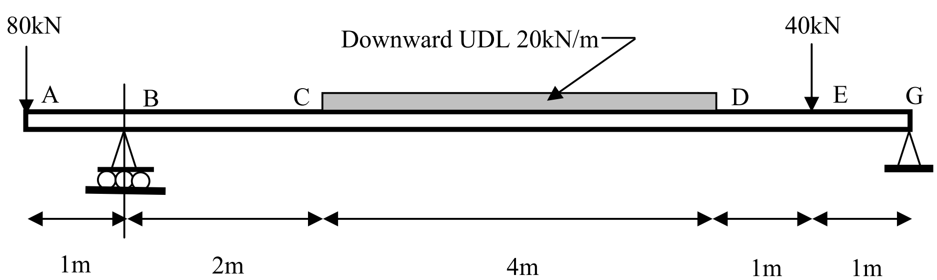
Consider the beam cross section shown.

Thickness 20mm.

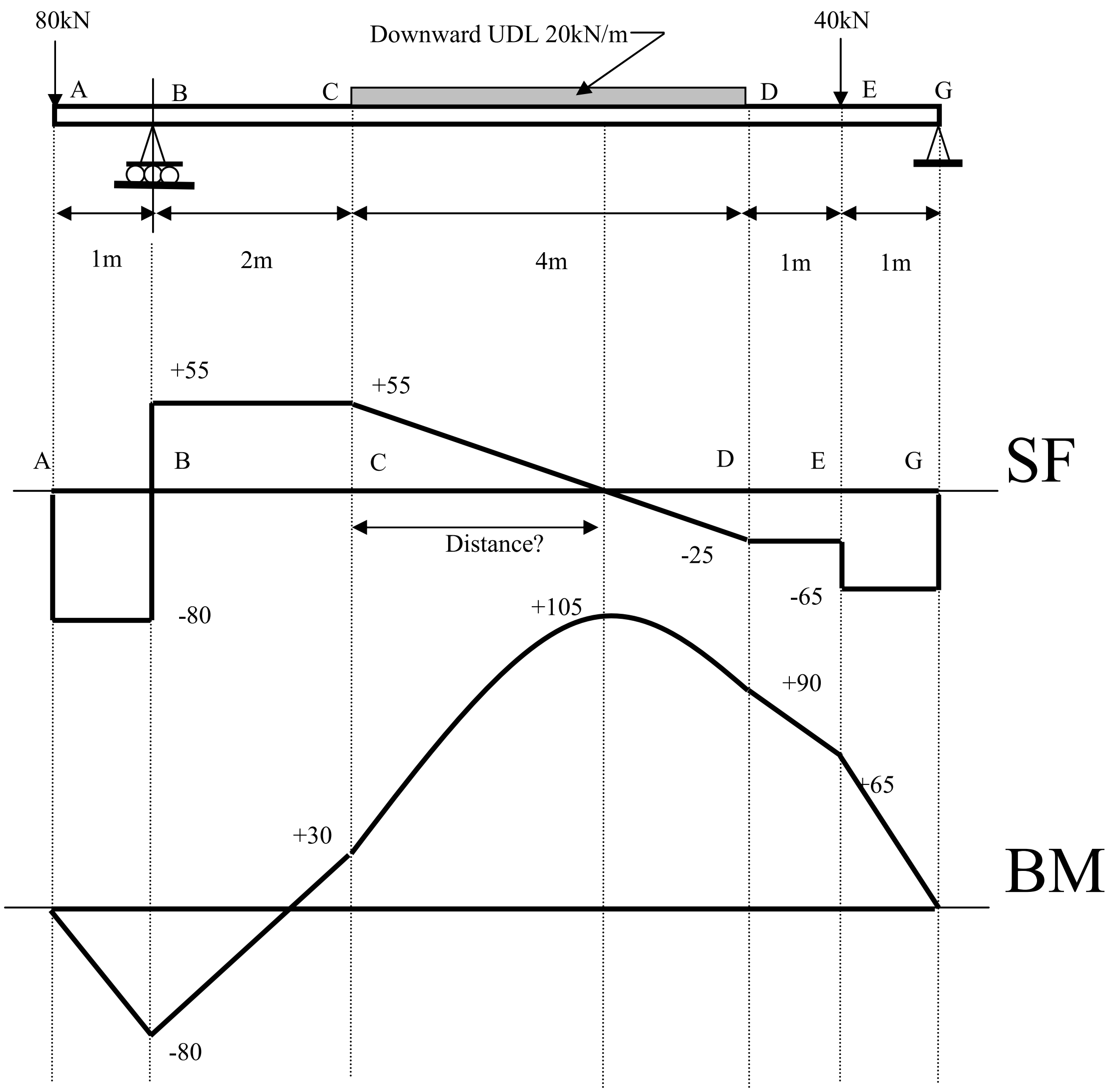


- Calculate the position of the centroid.
- Calculate I_{NA}

This beam section is then used in the configuration shown below:



- Determine the unknown reaction forces.
- Sketch Bending Moment and Shearing Force Diagrams for this beam.



BM equations:

B: -80x1 = -80kNm

C: -80x3 + 135x2 = 30KNm

H: -80x5.75 + 135x4.75 – (20x 2.75^2/2) = 105kNm

D: +65x2 – 40x1 = 90kNm

E: +65x1 = 65kNm

Before carrying out any stress calculations:

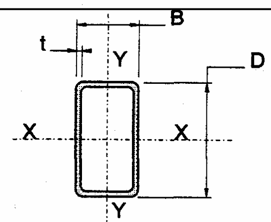
a) Can you predict the position of the maximum TENSILE stress?
b) Can you predict the position of the maximum COMPRESSIVE stress?
c) Decide which will have the greatest magnitude – why?
d) After you have made these predictions – calculate the maximum TENSILE and COMPRESSIVE stress values to check if you were right.

Tutorials

1. In the design of a wing spar, a beam with a hollow rectangular cross section is used. The beam is simply supported, 10m long and is under a UDL of 0.5 kN/m.
   Determine the beam dimensions from the data supplied below if a safety factor of 2 is applied and the material maximum stress is 250 MPa.

Solution stages:

- Establish the maximum bending moment in the beam using SF and BM analysis.
- Apply the factor of safety to the allowable stress.
- Find the minimum Section Modulus allowed.
- Select the appropriate beam cross section from the data (more than one is possible).



Designation Size D x B mm x mm	Thickness t mm	Mass per metre kg	Area of section cm^2	I_{XX} cm^4	I_{YY} cm^4	Z_{XX} cm^3	Z_{YY} cm^3
100 x 50	3	6.75	8.60	111	37.1	22.2	14.8
	3.2	7.18	9.14	117	39.1	23.5	15.6
	4.0	8.86	11.3	142	46.7	28.4	18.7
	5.0	10.9	13.9	170	55.1	34.0	22.0
	6.3	13.4	17.1	202	64.2	40.5	25.7
100 x 60	3.0	7.22	9.2	125	56.2	25.0	18.7
	3.2	8.59	10.9	147	65.4	29.3	21.8
	5.0	11.7	14.9	192	84.7	38.5	28.2
	6.3	14.4	18.4	230	99.9	46.0	33.3
120 x 60	3.6	9.72	12.4	230	76.9	38.3	25.6
	5.0	13.3	16.9	304	99.9	50.7	33.3
	6.3	16.4	20.9	366	118	61.0	39.4
120 x 80	5.0	14.8	18.9	370	195	61.7	48.8
	6.3	18.4	23.4	447	234	74.6	58.4
	8.0	22.9	29.1	537	278	89.5	69.4
	10.0	27.9	35.5	628	320	105	80.0
150 x 100	5.0	18.7	23.9	747	396	99.5	79.1
	6.3	23.3	29.7	910	479	121	95.9
	8.0	29.1	37.1	1106	577	147	115
	10.0	35.7	45.5	1312	678	175	136

One
Solution: 120x60x5.0mm section.

2. A bar 2m long, 30mm wide and 5mm thick is supported horizontally on knife edges 1.2 m apart so that the overhang at each end is 0.4m. A downward force of 15N is applied to each end of the bar, and the upward deflection midway between the supports is found to be 17mm. Calculate the modulus of elasticity of the material from which the bar is made. (Answer: 203.3 GPa).

3. Determine the maximum bending stress in a 4m length of 30mm diameter mild steel bar, simply supported at its ends and acted on only by gravity. The density of mild steel is 8000 kg/m^3. (Answer: 41.86 MPa).

4. Determine the maximum TENSION stress in examples 1-3 of the previous SF and BM lecture. Assume a hollow cross section of 100x60x5mm using tutorial No 1 Table.